Where the Error of the Factorised Cumulant Chain Lives: Exact Identities, Cost Engineering, and a Cubic-Birth κ_4-Pattern Programme for Deep ReLU MLPs

WhestBench Phase 2, algorithmic-contribution write-up. Team Hack2Publish (participant foyshal_sk). Best graded submission: #334870 (v56As_fs), adjusted score 3.6521e-9, raw final-layer MSE 1.950e-8. Best nomination-safe submission: #334869 (v56s_fs), 3.6723e-9. *Draft of 10 October 2026. Every number is either a graded leaderboard result (submission id given) or a measurement whose result file is named in §8. Items still open are marked [OPEN]. Unmeasured quantities are marked (estimate) or (untested). Claims are labelled derived (algebra), measured (run by us) or graded (official grader).*

Abstract

We start from the public, MIT-licensed factorised third-cumulant chain of team 504aldo ("V29", raw 2.13e-8 at 0.254 of the budget) and pursue two questions: where its error is made, and what an estimator must contain to remove that error at an affordable cost.

Contributions.

1. **Two exact identities that we ship.** A spherical-input identity turns the Gaussian input into a uniform-sphere input plus one exact scalar, $a_1 = 0.99975589$. It is free and graded −1.1 % raw (#331787).
2. **A parity theorem.** It explains why every "mean-field" closure of the deep third cumulant fails.
3. **Cubic curvature-birth identities.** These come with readout factors, verified to ≤ 1e-14.
4. **Cost engineering under a metered budget.** Our tools are dead-neuron compaction, basis-side Strassen, aliased Grams, and an operation budget derived from a residual-time grader failure. Together they remove 23 % of the metered FLOPs for +2.8 % raw. With the zero-cost accuracy fixes they take the score from 5.418e-9 to 4.078e-9.
5. **An error map built from $6{\cdot}10^7$-sample Monte-Carlo teachers.** The closures are exact given true inputs. The error is fresh input error that accumulates through depth. The single most valuable missing input is the **per-neuron pattern** of the κ_4 diagonal: the teacher's pattern at the chain's own level would take the raw MSE from 2.290e-8 to 1.551e-8, which is the leaders' raw range.
6. **A κ_4-pattern programme with graded gains.**
   - A zero-FLOP linear corrector (V55) graded −5.4 % (#334591).
   - A cubic Hermite-star feature built from the chain's own young legs (V56) graded a further −4.4 % (#334805), reaching 3.6521e-9 (#334870).
7. **A ceiling ladder that explains why every earlier κ_4 carrier stalled near correlation 0.45.**
   - Carriers whose births stop at second order in the Hermite expansion of the ReLU are capped at 0.45–0.56.
   - Cubic births over a 6-layer horizon reach 0.80–0.85.
   - An age-resolved ceiling (9 Oct) shows that ages 1–6 are jointly needed. Ages 1–4 give 0.68–0.74, ages 2–6 give 0.69–0.71, and ages 4–6 give only 0.37–0.41.
8. **A table of the more than 60 architecture and representation ideas we killed, with their ceiling numbers.** It includes outside-AI proposals and states which of them came with no measurement.

9. **Derived constraints on any better method.** The next design target is stated quantitatively.
10. **The κ_3 input and the κ_3-dressed C·T channel (9–10 Oct, oracle and component level; not graded).**
    - A joint oracle shows that the teacher's κ_3 input is the larger single lever: ×0.607 against ×0.686 for the κ_4 diagonal on MLP #0, and ×0.716 against ×0.885 on the synthetic net. Together the two give ×0.272.
    - A C·T readout built from the chain's own κ_3 legs, ct_i = 24 Σ_a g_a Ȳ_ai X_ai Q_ai, lifts the star's ceiling from 0.47–0.52 to 0.82–0.85. Used as a feature in a teacher-fitted (oracle) affine map, it takes the plain chain to ×0.773, which is about 75 % of the perfect-κ_4 gain.
    - An exact age-collapse identity, Γ_a = G_a + Ystep_a Γ_{a+1}, together with grouping into dense A/P anchors plus exact low-rank anchors, cuts the source contraction to two dense products per source and target, plus one shared backward recursion. Low-rank sketches keep only ×0.843–0.885. At an estimated 40 u this is about score-neutral, so the channel is real but not yet affordable. A rank-32 eigen-sketch (10 Oct) keeps ×0.874 at half the rank-64 thin-product count, an estimated 3–4 % score gain on the plain chain under the same oracle map (untested on V56).
11. **Where the κ_3 error is, and three mechanisms killed (10 Oct; oracle and component level; not graded).**
    - Measured against an estimate of the teacher's own sampling noise, the chain's early-layer κ_3 "error" is mostly teacher noise. The real (2,1)-slice error grows from about 3 % at L4 to about 7 % at L16, and the d3 level shrinks by 1.4 % at L16. The teacher's κ_3 helps only at L9–16 (§4.10).
    - A missing cubic-response C×κ_3 term, a feedback-scale confound and the $\kappa_4 \to \kappa_3$ feed channel were each tested and killed. The last kill holds only for the tested sample size, N = 10^6 (§6.5).
    - Local corrections of one channel repeatedly make the final mean worse; only joint input oracles give large gains (§7.5).

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1. Setting and baseline

1.1 Task and scoring

For a bias-free ReLU MLP of width n = 1024 and depth 16, with He-initialised weights and input x ∼ N(0, I), the estimator returns E[h^(l)] for every layer. The score is

score = MSE_final × max(0.1, FLOPs / 2^41).

The final-layer MSE is measured against a 10^9-sample Monte-Carlo reference. FLOPs are metered analytically by flopscope.

We use the unit **1 u = 2^31 FLOPs**, which is about one dense 1024^3 product, so the budget is B = 1024 u. "util" denotes FLOPs/B, and the 0.1 floor equals 102.4 u.

The grader enforces three hard limits per MLP:

- 120 s wall time;
- 0.4 s of **residual** Python time outside metered operations;
- an 8 GB address space.

A capped MLP receives zero predictions with multiplier 1, which scores about mean(target^2) ≈ 0.35. A single capped MLP in a hundred therefore ruins a submission.

1.2 The public chain (V29, 504aldo, MIT)

Our estimator is a modification of 504aldo's public "V29". We use it under the MIT licence; its copyright header and LICENSE are retained in every file we ship.

V29 implements the factorised K = 3 "simple" cumulant propagation of Wu et al. (arXiv:2605.05179, §4.3). Per layer it carries:

- the mean;
- the covariance C;
- the third cumulant κ_3, as a **sum of sources**.

Each ReLU injects one source. A source is stored as:

- a propagator leg P_s (n×n, the identity at birth, updated $P \leftarrow W^T \operatorname{diag}(w_1) P$);
- a birth leg A_s = P_s a_b;
- thin n×16 feedback legs.

The next ReLU needs the slices D3_i = κ_iii and D21_ic = κ_iic. These are obtained per source from contractions of Hadamard products of the legs.

Further features of V29:

- **Shared basis for old sources.** Sources older than four layers are confined to a shared basis of rank 384, and the oldest to a nested rank-224 sub-basis.
- **κ_4 handling.** κ_4 is not carried. It is regenerated each layer by a memoryless law, G_off = λ_l C_off.
- **D21 feedback.** D21 enters through feedback legs.
- **Pooled Strassen–Winograd.** Products use pooled, batched Strassen–Winograd at level 5.

The author's ledger prices V29 at 260 u, split as:

Component	Cost
Young sources	116 u
Old sources	107 u
Thin and elementwise	25 u
Covariance	7 u
Closure	6 u

The author also reports a tolerance law: extra final MSE ≈ $4.2\text{e-}6 \cdot \varepsilon^2$, where ε is the relative RMS error of D21. D21 must therefore be right to about 2 %.

Graded baseline. V29 verbatim graded 5.418e-9 (raw 2.133e-8, util 0.254; #331681), matching the author's own submission.

1.3 Measurement instruments

All of the following are described in §8.

- **Monte-Carlo teachers** with $6 \cdot 10^7$ samples on public mini MLP #0 (logan-fitzgerald) and on a synthetic 1024×16 He network. They record, per layer, the pre- and post-ReLU means, covariances, κ_3 slices and κ_4

diagonal and slices.

- **Oracles** that replace one internal quantity of the chain by the teacher's value and rerun the chain end to end.
- **A deterministic 100-MLP paired harness** on Kaggle, using the public mini split with its 10^9-sample truth.

The project κ_4 metric. Throughout §5, the metric is the correlation of the standardised κ_4 diagonal, κ_4/v^2, with the teacher's, computed on *decisive* neurons ($|\mu/\sigma| < 1.5$) at layers 12, 15 and 16. Raw (unstandardised) correlations inflate every candidate because they contain v^2, which the chain already gets right. One outside proposal scored 0.80–0.87 on raw correlation and 0.43 on this metric.

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2. Exact identities and derivations

2.1 Spherical-input identity (derived; shipped as V31)

Write $X = (R/\sqrt{n})\cdot Y$, where Y is uniform on the unit sphere scaled to radius $\sqrt{n}$ and $R \sim \chi_n$ is independent of Y.

A bias-free ReLU network h is positively homogeneous of degree 1, so

$E[h(X)] = E[R/\sqrt{n}] \cdot E[h(Y)] = a_1\, E[h(Y)]$, with $a_1 = \sqrt{(2/n)}\,\Gamma((n+1)/2) / \Gamma(n/2) = 0.99975589$ for $n = 1024$.

The sphere input has $C = I$ and $\kappa_3 = 0$. Its fourth cumulant is $\kappa_4 = c\cdot \mathrm{Sym}_3(I\otimes I)$, with $c = -2/(n+2)$. In the chain's harmonic-core representation this is a previous-layer core $g = -6/(n+2)$.

The estimator therefore does three things:

- it injects that core at layer 0;
- it runs the existing κ_4 machinery unchanged;
- it rescales the output by a_1.

Checks.

- **Measured:** the injection reproduces the identity's scale shift to 0.3 % (+2.450e-4 against +2.442e-4 predicted). The control without a_1 gives 8.02e-8.
- **Graded:** raw 2.132e-8 → 2.108e-8 (−1.1 %), score 5.137e-9 (#331787), at zero cost.

The reparameterisation idea comes from a public Phase-2 repository (barnobarno666), surfaced in an outside-AI consultation (§9). On that repository's K3 chain it was worth −7.9 %. On V29, most of that gain is already absorbed by the fitted κ_4 regeneration.

2.2 Parity theorem: no mean-field part of the deep κ_3 (derived)

Take a source born at layer b. At any later layer l its legs are $P = W^{T(l)}\, D\, P'$ and $A = P\, a_b$, where P', a_b and the birth coefficients do not depend on $W^{(l)}$.

- Every quantity that is linear in the source's κ_3 (the products H, T2, T3 and T4, and the diagonal D3) is therefore an **odd** function of $W^{(l)}$.
- $C^{(l)}$, the leg Grams and the per-neuron variances are **even** functions of $W^{(l)}$.

Because the weight ensemble is sign-symmetric,

$E[\, H_s^{(l)} \mid C^{(l)}, \text{Gram}(A_s, P_s), \text{all state before } l \,] = 0$, exactly.

Consequences.

1. Any closure that replaces old sources by a function of even statistics discards the entire contribution. This covers mean-field closures, scale mixtures and polynomials in C. The failures of such closures reported in the public log (9.1e-7 and 2.3e-6) are therefore structural.
2. Replacing a source by its conditional expectation given one fresh weight row has a relative-error floor of $\sqrt{(6/(r+8))}$ in the retained support dimension r. That is 12 % at r = 384 and 16 % at r = 224, an order of magnitude above the 2 % tolerance.
3. κ_4 terms built from two κ_3 factors are even, and $C \otimes C$ is the only even invariant available. This explains the empirical law $G_{off} = \lambda_l C_{off}$.

The argument originated in an LLM consultation (Fable 5) and was checked by us. The floor was derived independently by two other models with the same constant. A final written hand-verification is still listed as pending in our notes **[OPEN]**.

The learned-residual experiment in §4.7 is the data-side counterpart of this theorem.

2.3 Exact structure of the births (measured to rounding)

- **Birth leg.** The birth leg satisfies $A_s = P_s \cdot \text{diag}(w_1, b) \cdot C_{off}^{(b)}$ to 1e-5 at every slot and layer: the birth operator is the row-scaled off-diagonal post-ReLU covariance. It has a zero diagonal and a flat spectrum, needing 310–809 singular directions for 99 % of its energy, so it is not compressible.

- **Born tensor.** The born tensor splits into two families:

  - AAP, with $|AAP|/|total| = 0.36–0.37$;
  - MPP, with 0.74–0.75.

  Their mutual cosine is +0.56–0.57.

2.4 Edgeworth port deficit (derived and measured; shipped as FB 1.25)

Expanding the post-ReLU κ_3 to first order in C shows that the newborn $\text{Sym}(X_1 \otimes I \otimes Y_1)$ carries exactly **half** of both K×C all-distinct families.

Measurement. Regression coefficients on the slice residual are −0.55 to −0.69 and −0.6 to −0.9, against −1/2 predicted.

Restoring the missing families fully does not help, but a partial restoration does:

Restoration	Effect on the teacher pair
Full (S = 2)	+3.4 % / +3.1 % (worse)
Quarter (S = 1.25)	−0.8 % / −0.35 %

The fix is zero-FLOP. It was shipped together with an adaptive-λ exponent of 1.2. On the 100-MLP harness this scored ×0.9946 ± 0.0007, and it graded 5.1284e-9 (#332110). Why the full fix hurts the final mean even though it improves the slice at layers 4–10 is an open paradox **[OPEN]**. A 10 Oct test removed one explanation: the quadratic feedback cross term that also grows with the scale is inert, and the linear restoration itself is what hurts (§6.5).

2.5 Cubic curvature-birth cumulant identities (derived; numerically verified)

Expand the ReLU around a neuron's pre-activation mean μ and standard deviation σ, with α = μ/σ and standardised input u:

relu(μ + σu) = Σ_k a_k He_k(u), a_k = σ^k E[relu^(k)(μ+σu)] / k!.

Since relu″ = δ:

- $a_2 = \sigma\varphi(\alpha)/2$;
- $a_3 = -\sigma\alpha\varphi(\alpha)/6$.

The second coefficient a_2 is the "curvature" coefficient. The third coefficient a_3 is the cubic coefficient used by the Round-18 star and by V56, where it appears as c = −αφ/6 in standardised units.

For jointly Gaussian standardised u, Wick's theorem gives the cubic-birth identity

$\kappa(He_3(u_a), u_b, u_c, u_d) = \kappa(u_a^3, u_b, u_c, u_d) = 6\,\rho_{ab}\,\rho_{ac}\,\rho_{ad}$.

A cubic Hermite term at hub a thus injects a **star**-shaped fourth cumulant into all downstream neurons. Quadratic terms pair up into **path** (He_2–He_2) and C·T shapes.

Numerical verification (`r17_curvature_identities.json`):

- the cubic identity and its He_3 form hold to 8.1e-16 and 6.3e-16;
- the quadratic Q1 and Q2 identities hold to 5.1e-16 and 2.4e-16;
- the combinatorial readout factors hold to 9.0e-15 for the star (24), 3.5e-15 for C·T (24) and 4.1e-15 for the path (48).

These identities are the algebraic basis of §5.4–§5.5.

2.6 Counting results on the sum-of-sources family (derived or enumerated)

- **Young-tier floor.** An exact enumeration of contraction orders gives 193 square products for the young tier in every order: 39 P + 54 A + 100 D21 = 98.4 u at the measured 0.505 u per product. Two independent outside validators reproduced the count.
- **Duality.** Storing only birth matrices and layer maps and reading slices by reverse-mode differentiation (the "lazy cubic circuit") costs about 4l products at depth l, about 245 u in total, against 249 u for the source chain. The two are dual contraction orders of one tensor network.
- **Irreducibility.** Sym^3 is an irreducible GL(n) representation. A linear state closed under every dense M^⊗3 therefore needs C(n+2, 3) ≈ $1.8\cdot10^8$ numbers.
  - *Caveat:* with fixed, known weights, only what future readouts can observe matters, which is at most n^2 per remaining layer.
  - The statement is a count over known contraction orders, not a lower bound on arithmetic cost.
- **Algebraic floor per source.** Exactly evaluating one old source's D21 costs at least min($2n^2r$, $4nr^3$) per source-layer.

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3. Cost engineering under a metered, op-capped budget

3.1 Where the FLOPs go

A per-call-site ledger of metered FLOPs (`m1_matmul_ledger.py`) on the V29-era chain gives 249 u per network. Of this, 220 u (88 %) is matrix products.

Family	Cost
Young-source transport	≈ 54 u
Young-source D21 contractions	≈ 50 u
Joins (10 per network)	41–46 u
Old sources in the basis	≈ 41 u
Thin feedback	≈ 15 u
Strassen additions	≈ 12 u
Einsum	9 u
Covariance and closure	≈ 8 u

The young tier sits at its enumerated floor (§2.6), so cost had to come from elsewhere: joins, dead neurons, the basis side, and the meter's pricing rules.

3.2 Graded cost steps

Each step below is exact or near-exact. "Util" is the grader's figure, or adjusted ÷ raw where that is the only source.

Step (date)	Mechanism	Graded score (id)	Raw	Util
V29 verbatim (20 Sep)	—	5.418e-9 (#331681)	2.133e-8	0.254
STRASSEN_MIN 32→8	Two more Strassen levels on the rank-384/224 families. Prediction drift 4.9e-13, peak 5.97 GB	5.220e-9 (#331686)	2.132e-8	0.244
+ STRASSEN_FIRST 5	First call per worker at full depth	5.195e-9 (#331732)	2.132e-8	0.2436
+ V31 sphere (§2.1)	Accuracy, 0 FLOPs	5.137e-9 (#331787)	2.108e-8	0.2437
+ FB 1.25, β 1.2 (§2.4)	Accuracy, 0 FLOPs	5.1284e-9 (#332110)	2.1046e-8	—
Transport join (25 Sep)	Rebuild the shared basis at a join by transporting the previous basis (no range finder)	4.9519e-9 (#332257)	2.1788e-8	0.2273
+ AGE_OLD 5, R_FB 4	One more dense young age; feedback rank 16→4	4.8066e-9 (#332324)	2.1371e-8	0.2249
v48 compaction (26 Sep)	$w_1 := 0$ for $\Phi <$ 1e-2; live-row contraction, widths padded to multiples of 32	4.6294e-9 (#332391)	2.1462e-8	—
v49 basis-side Strassen + aliased Grams	Projections, rotations, Qc and the D21 lift through Strassen; joiner Grams as YY^T (billed at half)	4.4157e-9 (#332401)	2.1463e-8	—
v50 D21 live columns	D21 columns of dead neurons never computed	4.2512e-9 (#332404)	2.1537e-8	—
v51 D21 rows, $\Phi <$ 3e-3 / 7e-3	D21 rows of very dead neurons never computed	4.1081e-9 (#332418) / 4.0902e-9 (#332420)	2.1545e-8 / 2.1636e-8	0.189
v52 op budget, safe B / A (27 Sep)	Strassen only where it saves ≥ ∼2 u per 1,000 ops (§3.3)	4.1696e-9 (#332566) / 4.1228e-9 (#332567)	—	—
+ R_OLD 352, Tq from the QR R factor, R_FB 2	Exact or near-exact trims	B: 4.0995e-9 (#332626); A: 4.0784e-9 (#332621)	2.197e-8 (B)	—

Net effect. From V29 to these variants, the trims removed 23 % of the metered FLOPs (util 0.244 → 0.189) for +2.8 % raw (2.105e-8 → 2.164e-8). On 26 Sep alone, the score fell from 4.8066e-9 to 4.1081e-9 (−14.5 %).

What could not be compacted. Only the *input* side of dead neurons can be compacted. Each of the following loses, measured on the teacher pair:

Attempt	Effect
Removing dead neurons as hubs of the new source (their own κ_3 must propagate)	×6.40 / ×6.74
Zeroing their entries of the transported covariance	×1.59 / ×1.46
Zeroing rows with $\lvert\alpha\rvert > 3$	+6.5 % / +19.6 %
Treating always-on neurons as hubs	×10.0 / ×8.0

3.3 The residual-time cap and the operation budget (graded failures)

Two variants that ran cleanly on our machines graded catastrophically:

Submission	Grade	Share of MLPs capped
#332402 (v49 + R_OLD 320)	0.0434	≈ 12 %
#332425 (v51, row threshold 1e-2)	0.0204	≈ 6 %

Memory was not the cause: the local peak was 5.51 GiB. We measured that the residual Python time tracks the number of metered operations, with no other dependence on the variant:

- about 0.07 ms per op on our laptop;
- 0.053–0.059 ms per op on Kaggle, for every variant;
- about 0.02 ms per op inferred for the grader.

The failing variants issued 21.7k–22.3k operations; v48, at ≈ 16.4k, had passed.

The fix was an op budget.

- Strassen is used only where it saves at least about 2 u per 1,000 ops. Each top-level Strassen call costs 110–240 ops.
- The covariance slot rides inside the compacted family.
- The op ledger names the expensive groups:

Op group	Ops	Cost saved
Nested projections	1.3k	0.7 u
Qc	1.25k	1.7 u
D21 lift	1.2k	1.6 u
C slot	1.27k	0 u

Both classes graded with no capped MLP (#332566 and #332567). This produced our two submission classes:

- **B (nomination-safe):** ≤ ~18k warm-call ops.
- **A:** ~19.9k–20k ops.

3.4 The 8 GB limit (graded failures)

The first V56 implementation batched the cubic star as a 3-D broadcast. It failed the grader's smoke test four times (#334786, #334787, #334798, #334799).

Kaggle reproduced the failure only under `whest run --runner subprocess`, which gave "Unable to allocate 450. MiB for an array with shape (118013952,)". In-process runs passed, with maxrss 5371 MB against 5346 MB for the V55 base.

A bisection submission with the star disabled graded exactly as V55 (3.8546e-9, #334803). The per-slot 2-D-plus-mat-vec form then passed (#334805). Strassen level 6 was ruled out earlier for the same reason, with a peak of 8.13 GB.

Lesson. `whest validate` catches neither the residual cap nor the address-space limit. Subprocess runs on the mini split are the minimum pre-submission check.

3.5 The one-for-one law, and where it breaks

Inside the chain, every *accuracy knob* we swept on the 100-MLP harness trades about 1 % cost for 1 % accuracy, and the shipped constants sit at the score optimum:

Knob	Score ratio
β 1.2	×0.9993
λ ×0.9	×1.0026
Two tier-1 passes	×1.073
FB × β grid	within +0.12 %
R_OLD 448	×1.0113
AGE_OLD 6 on V56	raw ×0.978 but util +3.7 %, score ×1.014

The law does **not** hold for **cost levers that are exact or nearly exact**: compaction, Strassen placement, aliased Grams and the transport join. Our 27 Sep audit of 74 earlier kills found that all 8 kills that had to be revived were cost or knob levers. They had wrongly been killed on MSE alone, on mixed sub-populations, in one direction only, or on a different base.

No architecture kill was revived. Every one had missed its gate by a factor of 5 to 50.

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4. The error map: where the error lives

All results in this section are measured on MLP #0 against the $6{\cdot}10^7$-sample teacher, and partly replicated on the synthetic network. The plain chain's final MSE in these runs is 2.290e-8.

4.1 The closures are exact; the error is accumulated input error

- **Mean closure.** The Edgeworth mean closure, evaluated on the teacher's true marginals (μ, var, κ_{3iii}, κ_{4iiii}), matches the teacher's ReLU mean to 5e-11–1.5e-10 at every layer (9.3e-11 at L16).
- **Bivariate closure.** Teacher-forced at one layer, the bivariate covariance closure falls to the teacher's noise:

	L7	L14
RMS error per entry	7.04e-5 → 2.61e-5 (noise 2.5e-5)	8.84e-5 → 2.76e-5
Sensitivity-weighted next-layer variance error removed	87 %	91 %

- **Accumulation.** The chain's per-layer MSE grows smoothly, from 0.7e-9 at L1 to 2.29e-8 at L16, adding about 1–2.5e-9 per layer.
- **Gain.** The inherited mean error passes each layer with a gain of $|W\ \text{row}|^2 E[\Phi^2]$ = 0.65 at layer 2, rising to 0.8–0.96 at depth.
- **Inherited share.** Forcing the true mean at every layer takes the final MSE from 2.24e-8 to 4.2e-9, so about 80 % of the final error is inherited mean error.

4.2 Channel oracles

The true value of one channel is injected into the mean closure at every layer:

Exact input	Final MSE	Ratio
Variance	1.220e-8	0.533
κ_4 diagonal	1.609e-8 [1]	0.702
κ_3 diagonal	1.965e-8	0.858
Variance + κ_3	0.817e-8	0.357
Variance + κ_3 + κ_4	0.130e-8	0.057 (the truth's noise floor)
True κ_4 everywhere *except* the mean closure	2.311e-8	≈ 1
κ_4 diagonal + (2,2) + (3,1) slices	1.738e-8	0.759
(3,1) only / (2,2) only / both	2.353 / 2.796 / 2.680e-8	1.03 / 1.22 / 1.17

[1] Two runs give 1.600e-8 and 1.609e-8 (`k4oracle2_target_base.json`: 1.6086e-8). An independent replication by a subagent gave ratios 0.532 / 0.865 / 0.702 / 0.057.

Single-channel ratios are upper bounds on each channel's own share, because each includes error inherited from the other channels.

Two facts follow from the table:

- κ_4 acts **only** through the mean closure, and only through its **diagonal**.
- The off-diagonal κ_4 slices are harmful when injected alone.

A first-order attribution of the L16 error gives:

Source	Share
Pre-activation mean	1.84e-8 (80 %)
Variance	2.0e-9
κ_{3iii}	1.0e-9
κ_{4iiii}	1.0e-9

At depth, the *fresh* error per layer splits about 50 % variance, 25 % κ_3 and 25 % κ_4. The fitted coefficients are −0.99, −0.98 and −1.04, against −1 in theory, with R^2 0.93–0.97 from L9 on.

4.3 The chain's per-channel error against the teacher

Teacher noise is given in parentheses where it was measured.

Layer	Mean (σ units)	Variance	d3	κ_4 diagonal (slope, corr over all neurons)	C off-diagonal	D21 off-diagonal
7	1.9e-4	0.04 % (0.02)	3.2 % (1.1)	15 % (0.95, 0.78)	0.39 %	4.6 %
11	3.1e-4	0.09 %	3.8 %	22 % (0.89, 0.78)	0.59 %	5.7 %
15	4.6e-4	0.14 %	4.6 %	32 % (0.78, 0.83)	0.73 %	6.6 %
16	5.1e-4	0.14 %	4.7 %	34 % (0.75, 0.82)	—	—

Over all neurons the chain's κ_4 diagonal correlates 0.78–0.83 with the truth. On the **decisive** neurons that drive the mean closure it correlates only 0.26–0.34, at a level ratio of about 1.0. This gap between the two populations is why §5 uses the decisive-neuron metric.

4.4 Variance error is inherited covariance error

About 95 % of the variance error is $dvar_i = w_i^T \delta C\, w_i$, a random projection of the off-diagonal post-ReLU covariance error. That covariance error has two parts:

- an inherited part, with slope 1.01 and R^2 rising from 0.52 to 0.79 with depth;
- a fresh part of about 4.2e-5 RMS per entry.

The audited missing covariance terms explain little:

- the C^3, C^4 and $C\times\kappa_{31}$ patches have RMS 2–3e-6, against an error of 6.5–9e-5;
- the chain's own (1,1) terms explain only 3–8 %.

There is no missing covariance term. The variance channel's 47 % comes from accumulated κ_3 and κ_4 input error.

4.5 κ_3: the all-distinct entries, and memory

The chain's $\kappa_3(i,i,j)$ slice has slope 1.00 and R^2 0.996–0.998, but a random error that grows from 4.1 % at L3 to 6.7 % at L15. Of that error:

- 91–96 % sits in the **all-distinct** post-ReLU entries $\kappa_3(h_i,h_j,h_k)$, which are themselves 9–16 % wrong;
- it is not truncation: with no confinement the slice error stays at 6.6 % and the final MSE moves only −3.9 %, for +19 % cost;
- about 55 % of it is unexplained even at second order in K, since the twelve K×K terms have RMS 3e-6 to 2e-5 against an error of 2.2e-4.

Refinement (10 Oct, §4.10). These percentages are measured against the $6\cdot10^7$-sample teacher and include its sampling noise. A Gaussian estimate puts that noise at 2.5 % (L4) to 5.3 % (L2) on the slice, so most of the 4.1 % at L3 is teacher noise. Net of noise, the chain's slice error is about 3 % at L4 and grows to about 7 % at L16.

Perfect κ_3 oracle. A perfect κ_3 input (slice + d3) is worth ×0.6075 on MLP #0 (1.382e-8 vs 2.275e-8) and ×0.715 on the synthetic net. The synthetic figure is partial: 8 of 14 layers. Applied in layer windows on MLP #0 / synthetic:

Window	Ratio
Layers ≥ 8	0.637 / 0.754
Layers ≥ 12	0.704 / 0.863
Layers ≤ 11	0.884 / 0.837

The windows compose multiplicatively: 0.884 × 0.704 ≈ 0.62.

Memory. The κ_3 memory is long and coherent but not low-rank in age. Keeping only sources of age ≤ h leaves this relative D21 error at L15:

h	2	4	6	8	10
Error	84 %	65 %	46 %	31 %	16 %

- The age contributions are coherent: mean pairwise cosine 0.29–0.39.
- They are not low-rank: projecting on six age-modes still leaves 8.5 %.

- Removing the old tier entirely costs 32× (7.43e-7). Every age gate follows MSE ≈ 7.4e-7 × 0.66^(g−4).

4.6 κ_4: level transported, pattern regenerated

- **One-step transport.** Pairwise transport of the true post-ReLU slices gets the next layer's κ_4 diagonal **level** right (slope 1.00) but not its **pattern** (corr 0.24–0.59).
- **Exact slices are not enough.** Even with the exact (4), (3,1) and (2,2) slices, the next diagonal is 14 / 22 / 33 / 43 / 46 % wrong at L4 / L8 / L12 / L15 / L16.
- **Where the content lives.** The next layer's diagonal is about 90 % the (2,2) pair term. The next d3 comes 90–99 % from off-diagonal κ_3, so the diagonal-to-diagonal channel carries only 18 % (L4) down to 1.7 % (L16) in norm.

4.7 What does not carry the missing information

Gaussian and copula closures. The one-step closure floors (2e8 samples per layer) give these RMS errors of the next-layer mean:

	L4	L8	L12	L15
Gaussian closure	5.2e-4	5.1e-4	4.6e-4	4.5e-4
Gaussian copula with exact marginals	1.14e-3	9.2e-4	8.1e-4	7.5e-4

The Gaussian closure errs at about 3× V29's final error. The copula with exact marginals is worse than the plain Gaussian closure at every layer.

Learned residual correctors. On 100 MLPs (102,400 output neurons), learned correctors recover almost nothing:

Model	Residual MSE recovered
Ridge, 19 features	0.3 %
Ridge, 189-feature quadratic expansion	0.7 %
Gradient boosting	0.5 %

- Per-network offsets explain 0.44 % of the residual variance.
- Every single feature correlates ≤ 0.04 with the residual.
- This is the parity theorem (§2.2) seen from the data side.

Monte-Carlo routes.

- A 10^4-sample forward pass costs 156 u and delivers the variance only to 1.4 %, against a 0.1 % need.
- Control-variate surrogates decorrelate with depth:
  - the tangent network falls to 0.15–0.29 at L16;
  - the second-order Taylor surrogate falls from 0.967 at L2 to 0.00 at L16.
- The resulting variance reductions are only 1.0–2.2×.

4.8 The κ_3 and κ_4 inputs jointly (joint oracle, measured 9 Oct)

The oracle of §4.2 and §4.5 is rerun with both inputs at once (`experiments/variants/r20_joint_oracle.py`). The teacher's off-diagonal κ_3 (2,1) slice replaces the chain's D21 everywhere after it is computed, in the

closure terms and in the feedback legs. The teacher's d3 = κ_{3iii} replaces the chain's D3, and the teacher's κ_4 diagonal enters the mean closure.

The base is the plain V40-family chain (FB 1.25, β 1.2, spherical identity off) at util 0.2476. Its final MSE is 2.273e-8 on MLP #0 and 1.920e-8 on the synthetic net, slightly different from the 2.290e-8 base used in §4.1–§4.7. The synthetic teacher covers layers 2, 4, …, 14 and 15 only; at its other layers the chain keeps its own values, so the synthetic column is partial.

Teacher input injected (every layer unless noted)	MLP #0 final MSE	Ratio	Synthetic final MSE	Ratio
None (plain chain)	2.273e-8	1	1.920e-8	1
κ_3 slice + d3	1.380e-8	0.607	1.376e-8	0.716
κ_4 diagonal	1.561e-8	0.686	1.700e-8	0.885
d3 only + κ_4 diagonal	1.204e-8	0.530	1.394e-8	0.726
κ_3 slice (chain's own d3) + κ_4 diagonal	0.995e-8	0.438	1.429e-8	0.744
κ_3 slice (chain's own d3) + κ_4 pattern at the chain's level	0.977e-8	0.430	1.441e-8	0.751
κ_3 slice + d3 + κ_4 diagonal	**0.619e-8**	**0.272**	**1.092e-8**	**0.569**
κ_3 slice + d3 + κ_4 pattern at the chain's level	0.669e-8	0.294	1.163e-8	0.606
κ_3 slice + d3 at every layer, κ_4 diagonal from L8	0.653e-8	0.287	1.053e-8	0.548
Both inputs from L8 only	0.746e-8	0.328	1.102e-8	0.574

Reading the table:

- **κ_3 is the larger single lever on both networks:** ×0.607 against ×0.686 on MLP #0, and ×0.716 against ×0.885 on the synthetic net.
- **The two inputs compound more than multiplicatively.** The product of the single-input ratios would be 0.416 on MLP #0 and 0.634 on the synthetic net; jointly they give 0.272 and 0.569. Each channel's error is partly inherited from the other (§4.2).
- **With the true κ_4, the true d3 matters.** On MLP #0, adding the teacher's d3 to the teacher's slice takes 0.438 to 0.272.
- **The κ_4 pattern is again enough.** The teacher's pattern at the chain's level gives 0.294 against 0.272 for the full diagonal (compare §5.1). Injecting κ_4 only from L8 loses little (0.287).
- **What remains is mostly the variance channel.** After κ_3 and κ_4, the remaining ×0.27 is consistent with §4.2, where adding the true variance takes the joint ratio to the noise floor (0.057), and with §4.4, where the variance error is inherited covariance error.

4.9 Which Hermite order each channel needs, one step at a time (measured 9 Oct)

`r20_onestep_channels.py` runs $2{\cdot}10^6$ exact samples of z_l through MLP #0. At one layer l, the ReLU is replaced by its Hermite expansion of order K around the teacher's (μ, σ) (§2.5), with common random numbers, and z_{l+1} = act(z_l) $W^{(l)}$. Every next-layer statistic is compared with the **exact** ReLU step on the same samples. A chain whose births stop at order K makes at least this error in each channel per step.

Values are given at source layer L2 → L15 (layers 2, 4, 8, 12 and 15 were measured; the trend is monotone unless a range is given):

Order K	C off-diagonal	Variance	D21 off-diagonal	κ_3 diagonal	κ_4 pattern corr	κ_4 level ratio
1 (linear)	13.4 → 6.4 %	19.0 → 4.9 %	76.8 → 19.7 %	75.0 → 16.1 %	0.63 → 0.84	0.38 → 0.87
2	2.17 → 0.42 %	3.26 → 0.99 %	20.1 → 9.9 %	21.2 → 10.0 %	0.85–0.90	1.95 → 1.43
3	1.49 → 0.32 %	2.14 → 0.63 %	8.85 → 2.05 %	9.26 → 2.05 %	0.95 → 0.99	1.52 → 1.10
4	0.88 → 0.20 %	1.29 → 0.36 %	4.61 → 1.34 %	4.30 → 1.38 %	0.98 → 0.999	1.16 → 1.06

Errors are relative Frobenius errors against the exact step; the κ_4 columns use the project metric. For scale, this run's own distance to the $6{\cdot}10^7$-sample teacher is 1.5 % → 0.53 % (C off-diagonal) and 0.10 % (variance). The common random numbers make the K-against-exact differences much more precise than that.

Findings:

1. **κ_3 needs at least the cubic response.** With quadratic births (K = 2), the one-step D21 and d3 errors are 10–21 %, against a D21 tolerance of about 2 % (§1.2). K = 3 reaches 2.1–2.7 % only at L12–L15 and is 7–9 % at L2–L4. K = 4 is 1.3–4.6 %.
2. **The covariance needs at least the quadratic response.** The linear step alone leaves 6–13 % of the off-diagonal covariance.
3. **One κ_4 step is easy; compounding is not.** K = 2 gets the one-step κ_4 pattern to 0.85–0.90 but overshoots the level by 1.4–1.9×. Compounded over six layers, the same truncation falls to 0.45–0.55 (§5.4, set A), while K = 3 holds.

4.10 Where the κ_3 error is real, and where its gain lives (measured 10 Oct)

Error profile against teacher noise. `chain_k3_error_profile.json` compares the plain chain's κ_3 (the captures used in §5.8) with the $6{\cdot}10^7$-sample teacher at every layer. `teacher_k3_noise_estimate.json` gives a Gaussian approximation of the teacher's own sampling noise on the same quantities (an estimate). The "net" columns subtract that noise in quadrature, $\sqrt{(\text{error}^2 - \text{noise}^2)}$. This step is derived and assumes the noise is independent of the chain's error.

Layer	D21 off-diagonal error	Teacher noise (est.)	Net D21 error	d3 error	d3 noise (est.)	Net d3 error	d3 norm ratio, chain/teacher
2	4.4 %	5.3 %	≈ 0	3.3 %	5.4 %	≈ 0	1.002
4	4.2 %	2.5 %	3.4 %	2.9 %	2.2 %	1.9 %	0.998
8	4.9 %	1.2 %	4.8 %	3.1 %	0.9 %	3.0 %	0.996
12	5.9 %	0.7 %	5.9 %	3.8 %	0.6 %	3.8 %	0.993
16	6.9 %	0.5 %	6.9 %	4.6 %	0.4 %	4.6 %	0.986

Errors are relative Frobenius errors against the teacher.

Reading the table:

- **The early "error" is mostly teacher noise.** At L2 the estimated noise exceeds the measured difference on both the slice and d3.
- **The real slice error is carried and grows with depth**, from about 3 % at L4 to about 7 % at L16. At L7, L11 and L15 the profile gives 4.6 / 5.7 / 6.6 %, the same values as §4.3.
- **d3 has a level bias, not only scatter.** Its decisive-neuron correlation stays at 0.9965–0.9994, but its norm ratio falls steadily from 1.002 (L2) to 0.986 (L16), a 1.4 % shrink.

Where the κ_3 gain lives (`results/summaries/oct10/k3loc_{1,2,3}.json`). The teacher's κ_3 is injected only in a layer window, with the oracle of §4.8, on MLP #0. "Slice" is the off-diagonal (2,1) slice, "d3" the diagonal; "feedback legs" injects the slice into the feedback legs only. The base is the plain chain of §4.8 (2.2732e-8).

Teacher κ_3 injected	Window	Final MSE	Ratio
Slice + d3 (§4.8)	L1–16	1.380e-8	0.607
Slice + d3	L1–4	2.282e-8	1.004
Slice + d3	L5–8	2.141e-8	0.942
Slice + d3	L9–12	1.949e-8	0.857
Slice + d3	L13–16	1.832e-8	0.806
Slice only	L1–8	2.129e-8	0.937
Slice only	L9–16	1.897e-8	0.834
d3 only	L1–8	2.300e-8	1.012
d3 only	L9–16	1.938e-8	0.853
Feedback legs only	L1–16	2.276e-8	1.001

Reading the table:

- **The gain is in L9–16.** The product of the four slice + d3 windows is 0.653 (derived), against 0.607 when every layer is injected at once. L9–16 alone would give about 0.857 × 0.806 ≈ 0.69 (derived; not run as one window).
- **Early injection does not help.** Slice + d3 at L1–4 gives ×1.004, and d3 alone at L1–8 gives ×1.012. This fits the profile above: at L2–L4 the teacher's κ_3 is about as noisy as the chain's is wrong.
- **At depth, the slice and d3 matter about equally**: ×0.834 and ×0.853 at L9–16.
- **The feedback legs are not the lever**: ×1.001 (compare ×0.9999 in §6.3).

A κ_3 fix must therefore remove a carried late-layer error: 5–7 % in the slice and a 1.4 % shrink of the d3 level at L16. The three candidate mechanisms tested on 10 Oct are reported in §6.5.

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5. The κ_4-diagonal pattern programme

5.1 The target is the pattern, not the level

Oracle (`k4_oracle_run2.py`): the teacher's κ_4 diagonal is injected into the mean closure of the full chain. "Share" is measured against the full gain, none (2.290e-8) → diagonal (1.609e-8).

Injected target	Final MSE	Share of the gain
Teacher **pattern** at the chain's level	**1.551e-8**	**108 %**
Teacher pattern, layers ≥ 8 only	1.707e-8	86 %
Teacher pattern, layers ≥ 12 only	1.813e-8	70 %
Teacher pattern, layers ≤ 11 only	2.064e-8	33 %
50 % blend of teacher and chain patterns	1.789e-8	74 %
Chain pattern at the teacher's **level**	2.783e-8	worse than none (+21.5 %)

The whole κ_4 gap is the per-neuron *pattern* of the deep diagonal. The level the chain already produces is adequate.

Joint oracle: cheap κ_3 plus true κ_4. The same oracle on cheaper κ_3 chains shows that the gain sits on the full κ_3 chain only:

κ_3 configuration	Util	Own κ_4	True κ_4 diagonal	Ratio
Full chain	0.2476	2.290e-8	1.609e-8	×0.702
AGE_OLD 2	0.2303	5.240e-8	4.590e-8	×0.876
AGE_OLD 2, R_OLD 256/128	0.1784	1.288e-7	1.244e-7	×0.966
AGE_OLD 1, R_OLD 192/96	0.1381	6.511e-7	6.665e-7	×1.024

The route "cut κ_3 cost, recover with κ_4" is therefore closed. A κ_4 improvement must ride on the full κ_3 chain.

5.2 The deep pattern is carried memory

One-step carry of the true state (`opus12_k4_carry.json`, project metric):

Carrier	L12	L15	L16
Chain	0.339	0.264	0.324
Exact linear carry of the true κ_4	0.771	0.785	0.797
Linear carry + quadratic ReLU curvature	0.820	0.855	0.863

Gaussian physics alone is not enough. A Gaussian input with the true mean and covariance gives only 1–4 % of the deep κ_4 level from L8 on.

Memory horizon. Non-Gaussian history older than h layers is erased by a Gaussian restart, and the true network is run forward from there (`q18_k4_horizon.json`, N = 1e6). Pattern correlation at L12 / L16:

h	1	2	3	4	6
L12 / L16	0.23 / 0.20	0.40 / 0.41	0.53 / 0.59	0.67 / 0.65	0.78 / 0.77

Correcting for the teacher's noise raises h = 6 to 0.85 and h = 2 to 0.52. The later N = 4e6 runs (§5.4, set B) give somewhat higher values at h = 3–4, which is consistent with lower noise attenuation.

A young carrier spanning 2–3 layers is therefore dead from the start. An ideal one, tested out of sample, moves the G4-weighted error only from 0.210 to 0.203.

5.3 V55: a zero-FLOP linear pattern corrector (graded)

Construction. At every layer ≥ LMIN, a fixed linear map turns per-neuron quantities the chain already holds into a predicted pattern of κ_4/v^2. The inputs are:

- s3sq = $(D3/v\text{^}1.5)^2$;
- the row and column energies of the standardised D21;
- the correlation-row energy of C_pre;
- $|\mu/\sigma|$;
- the chain's own κ_4/v^2;
- a constant.

The pattern is rescaled to the chain's layer-mean level and fed to the **mean closure only**; the pair closure keeps the chain's κ_4.

The map is fitted once, offline, against the teacher on decisive neurons. The last layer uses a reduced map. The cost is 0 u and +310 metered ops (17,961 in total for the B class).

Pattern correlation.

- MLP #0: 0.41–0.55, against 0.24–0.41 for the chain (layers 6–15).
- Held out on the synthetic network: 0.41–0.52, against 0.25–0.35.
- The map transfers between networks: fitted on the synthetic net, it gives 0.43–0.56 on MLP #0.

Teacher pair.

Class	MLP #0 (fit network)	Synthetic net (held out)
B	2.1962e-8 → 2.0381e-8 (−7.2 %)	1.8856e-8 → 1.8343e-8 (−2.7 %)
A	−7.0 %	−2.7 %

Harness and grader. On 100 MLPs: ×0.9455 [90 % CI 0.9410, 0.9498], with 97/100 MLPs improved.

Class	Projected grade	Actual grade
B	3.8764e-9	3.8762e-9 (#334588)
A	3.8562e-9	3.8582e-9 (#334591)

Each is −5.4 % against its base. The s3sq coefficient ×1.25 then graded 3.8539e-9 (#334762) and 3.8331e-9 (#334763).

Sweeps place the map at its optimum. Ratios are on the 100-MLP harness.

Change	Ratio
s3sq ×1.5 / ×1.75 / ×2.0	×1.0068 / ×1.0207 / ×1.0396
c2 ×1.5 / ×0	×1.0563 / ×1.0131
Chain coefficient ×0.9–×1.2	within ×1.0011
Last-layer corrector off	×1.0116
Lever-weighted "lean" fit	×0.9957 against the decisive-neuron map, i.e. the lean fit is worse

Saturation. Squares, products and logs of the features add only +0.02–0.03 in sample. Previous-layer features carried by $(W \circ W)^T$ add nothing held out.

The cause is the feature set. With the same layer's teacher quantities, a least-squares combination reaches 0.65–0.70 out of layer, but almost all of that comes from the k22 row sum, a teacher-only feature that correlates 0.62–0.70 on its own. Without k22, out-of-layer fits fall to 0.39–0.54. **Per-layer chain features saturate near 0.5.**

5.4 Why carriers stalled: the Hermite-family ceilings

Between 27 Sep and 8 Oct we tested many κ_4 *carriers*: states transported from layer to layer with births at each ReLU. Every one stalled. Pattern correlation at L12 / L15 / L16:

Carrier (birth content)	Correlation	Source
Skew-lift Σ_s Sym4(T_s⊗r_s)	0.03 (best LSQR fit)	R13
Dense pair core born at the first ReLU	−0.35 to −0.22	R13
Signed bilinear Sym(H⊗C), H carried by W^THW	0.351 / 0.321 / 0.371	R16
Diagonal-only version of the same	0.336 / 0.294 / 0.351	R16
True (4) + (3,1) + (2,2) slices carried one step	0.30–0.45	R16
Pair-innovation births, all ages	0.433 / 0.362 / 0.411	R16b
Hub-pair C·T curvature births, h = 6	0.442 / 0.432 / 0.431	R17
Gaussian hypertree bank (GHQ-6)	0.559 / 0.471 / 0.499	R17

The pair-innovation carrier also made the closure error ×1.12–1.43 worse than the chain.

To find out why, we measured the ceiling of entire *families* of birth laws (`r17_hermite_family_ceiling.py`, Kaggle GPU, $N = 4 \cdot 10^6$). The construction:

1. Start from a Gaussian restart N(μ, C) of the teacher h layers back.
2. At each ReLU, replace the ReLU by its Hermite expansion of order K around the teacher's (μ, σ).
3. Measure the resulting κ_4 pattern.

Set A and set B: which Hermite orders are needed.

Mode	L12	L15	L16
Sanity: true ReLU, no restart	0.967	0.973	0.980
h = 6, true ReLU (restart ceiling)	0.859	0.820	0.818
h = 6, K = 2 (quadratic births)	0.548	0.449	0.530
h = 6, orders 0/1/3 only	0.511	0.498	0.469
h = 6, K = 3 (cubic births)	**0.848**	**0.799**	**0.817**
h = 6, K = 4	0.865	0.816	0.822
h = 4, K = 3 / true ReLU	0.749 / 0.753	0.687 / 0.702	0.711 / 0.710
h = 3, K = 3 / true ReLU	0.652 / 0.663	0.588 / 0.600	0.643 / 0.647
h = 2, K = 3 / true ReLU	0.517 / 0.525	0.436 / 0.434	0.485 / 0.494
True state 1 back, K = 2	0.850	0.875	0.881
True state 3 back, K = 2	0.687	0.689	0.738
True state 6 back, K = 2	0.527	0.477	0.578
True state 6 back, K = 3	0.915	0.927	0.949

Findings:

1. **Quadratic births are capped near 0.45–0.55.** This is exactly the band every second-order carrier landed in.
2. **K = 3 is necessary and sufficient.** K = 3 tracks the true ReLU at every horizon, and K = 4 adds ≤ 0.02.
3. **Both pieces are needed.** Dropping the quadratic term loses as much as dropping the cubic one.
4. **Compounding K = 2 diverges.** Applied over all 16 layers, it blows the level up by ×1e12. Even from the true state, quadratic births lose the pattern step by step: 0.85–0.88 after one step, 0.69–0.74 after three

and 0.48–0.58 after six (one-step channel errors in §4.9).

5. **Horizon.** The 0.70 gate needs a horizon of about 5–6 layers.

Set C: which ages need cubic births (9 Oct, `r17_hermite_family_ceiling_C.json`). The restart is at h = 6. The full cubic response is applied only at the listed ages, and the response is linear elsewhere.

Ages with cubic births	L12	L15	L16	Level ratio L12 / L15 / L16
1–6 (all)	0.848	0.799	0.817	0.81 / 0.53 / 0.48
2–6	0.709	0.708	0.691	0.61 / 0.44 / 0.35
1–4	0.736	0.682	0.713	0.28 / 0.19 / 0.19
1–3	0.647	0.582	0.637	0.13 / 0.09 / 0.09
1–2	0.525	0.458	0.456	0.05 / 0.03 / 0.04
1	0.343	0.239	0.231	0.02 / 0.00 / 0.02
4–6	0.367	0.397	0.409	0.30 / 0.19 / 0.20
6	0.026	0.116	0.187	0.05 / 0.04 / 0.03

Reading the table:

- **Ages are complementary.** No single age carries the pattern, and no late window does either: ages 4–6 give 0.37–0.41.
- **Dropping age 1** costs 0.09–0.14.
- **Truncating to ages 1–4** costs 0.10–0.12. Even so, it stays at or above the 0.70 gate at L12 and L16 (0.736, 0.713) and just below it at L15 (0.682).
- **Ages 1–3** fall to 0.58–0.65.
- **The level is a separate matter.** The level is mostly recovered from older ages (ages 4–6 alone give 0.19–0.30 of it). Since V55 and V56 rescale to the chain's level, and the pattern alone is worth 108 % of the oracle gain (§5.1), the pattern is the quantity a carrier must deliver.
- **Variance ratios** run from 0.64 (age 1 only) to 0.96 (all ages).

5.5 HTQB star → V56 (graded)

The Round-18 proposal "Hermite Tree Quartic Bank" (HTQB) uses the identities of §2.5 to read κ_4 from existing legs without a carried κ_4 state. It has two parts:

- a cubic **star** (Cu1: 24 Σ_a c_a P X^3, with $c_a = -\alpha_a\varphi_a/6$, over the last 6 ages);
- a quadratic **path** (Q2: 48 V^T C V, over 3 ages).

Its proposer priced it at 27.2 u.

Ceilings with teacher μ and C (`round18_htqb.json`), at L12 / L15 / L16:

Form	Correlation
Star + path, h = 6	0.637 / 0.559 / 0.586
Shippable form (star h6 + path h3)	0.616 / 0.538 / 0.541
Star alone	0.521 / 0.489 / 0.471
Path alone	0.074 / 0.024 / 0.150

Alone, HTQB is borderline. It is, however, **complementary** to the V55 features. With leave-one-layer-out fits on MLP #0, still using teacher μ and C (`r18_htqb_combo.json`):

Features	Correlation, L7–L15
V55 feature set alone	0.48–0.56
+ star	0.65–0.74 (0.70–0.74 at L12–15)
+ star + path	0.71–0.76 (L12 0.761, L15 0.711)

V56 ships the zero-FLOP half. The cubic star is computed from the chain's **own** young dense source legs (ages ≤ 6) and the per-source coefficient $c = -\alpha\varphi/(6 \cdot \text{var})$ recorded at birth. It enters as an 8th feature of the V55 map. The path is not shipped.

The 8-feature map reaches 0.53–0.71 per layer on MLP #0 and 0.54–0.66 on the synthetic net, depending on the fitting network (`k4s_fit.json`), against 0.41–0.55 for the 7-feature map. Utilisation is 0.1857, against 0.1855 for the V55 reference.

V56 grades.

Submission	Class	Score	Raw	Note
#334805 v56s_loop	B	3.6829e-9	1.9706e-8	−4.4 % vs the best B
#334811 v56As_loop	A	3.6597e-9	1.9539e-8	
#334869 v56s_fs (synthetic-fit map)	B	3.6723e-9	—	−0.29 %
#334870 v56As_fs	A	**3.6521e-9**	1.950e-8	−0.21 %, current best
#334871 fs + star ×1.25	B	3.6724e-9	—	no extra gain

Tuning is closed (`kaggle_v56_tuning.txt`). Ratios are against v56_ref (raw 2.0409e-8):

Change	Ratio
Star ×0.75 / ×1.5	×1.0110 / ×1.0111 (score)
Star ages 4	×1.0045
s3sq ×1.1 / ×1.25	×1.0028 / ×1.0093
AGE_OLD 6	raw ×0.9782 but score ×1.0138

The AGE_OLD 6 result is consistent with §5.4: a sixth dense age helps accuracy, but not at its present price.

5.6 Can Hermite trees replace the κ_3 chain? (measured; no)

Since the tree κ_3 diagonal has pattern correlation 0.975–0.986 with the truth, we tested whether a Hermite-tree estimator could drop the source chain altogether. The test is teacher-forced, one Edgeworth closure per layer (`r19_tree_ceiling.json`):

Layer	Teacher floor	Chain	Tree κ_3 + κ_4 (h = 6)	Chain κ_3 + teacher κ_4	Chain κ_3 + tree κ_4
8	4.3e-11	1.445e-9	2.99e-8	6.6e-10	1.08e-8
12	5.2e-11	2.004e-9	3.40e-8	9.5e-10	1.91e-8
16	6.2e-11	2.072e-9	4.18e-8	1.03e-9	1.93e-8

The tree is 10–20× worse. An oracle level rescale does not rescue it: L12 still gives 2.80e-8. The closure needs κ_3 to about 1 %, and a 0.98 correlation is far from that. The tree κ_4 is also harmful as a direct replacement of the chain's κ_4; it is useful only as a *feature* inside the rescaled pattern map.

5.7 Summary of the graded κ_4 gains

Stage	Score (id)
Before V55 (A)	4.0784e-9 (#332621)
V55	3.8582e-9 (#334591)
V55, s3sq ×1.25	3.8331e-9 (#334763)
V56	3.6597e-9 (#334811)
V56, synthetic-fit map	**3.6521e-9 (#334870)**

In total this is −10.5 % on the score at unchanged cost, all of it from making the κ_4-diagonal *pattern* better at zero FLOPs.

5.8 Beyond the star: the κ_3-dressed C·T channel (measured 9–10 Oct; oracle and component level, not graded)

The V56 star is the *Gaussian* cubic birth. The Round-17 hub-pair C·T births were killed alone at 0.43–0.44 (§5.4). A design from a 9 Oct consultation, the "lagged κ_3-dressed vertex" (LKV), reads the chain's κ_3 through the quadratic and cubic Hermite vertices of §2.5. Every number below is our own measurement on public MLP #0.

Readouts. Take a vertex layer m and a target pre-activation layer t, at age j = t − m. Write u = z_m − μ_m and the K = 3 response of the ReLU at m as

$h - E\,h = \Phi\,u + g\,(u^2 - v) + c\,(u^3 - 3\,v\,u)$, $g = \varphi/(2\sigma)$, $c = -\alpha\varphi/(6\sigma^2)$.

With the one-step linear response Ystep_l = diag($w_1^{(l)}$) $W^{(l)}$ (legs move as L ← Ystep_l^T L, §1.2), define:

- the transports $\dot{Y}$ = Ystep_m ⋯ Ystep_{t−1} and $\bar{Y}$ = $W^{(m)}$ Ystep_{m+1} ⋯ Ystep_{t−1} (the hub leg, with no w_1 on the hub);
- the C-leg X = C_m $\dot{Y}$;
- the κ_3 object Q_ai = Σ_bc κ_3(z_m)_abc $\dot{Y}$_bi $\dot{Y}$_ci.

The single-vertex births then read out on the diagonal at t (readout factors from §2.5):

star_i = 24 Σ_a c_a $\bar{Y}$_ai X_ai^3 (the Gaussian cubic star, shipped in V56) ct_i = 24 Σ_a g_a $\bar{Y}$_ai X_ai Q_ai (κ_3 converted at a quadratic vertex: the C·T channel) tt_i = 36 Σ_a c_a $\bar{Y}$_ai YL_ai Q_ai, YL = (D21 + diag d3) $\dot{Y}$ (κ_3 squared at a cubic vertex: T·T)

The prediction sums the readouts over ages j = 1–6.

Ceiling (`fable_ceiling_2026-10-09.py`, $2{\cdot}10^6$ samples through the true network). Q, X and the exact births are computed from the samples. Project metric at L12 / L15 / L16:

Readout, ages 1–6	Pattern correlation	Level ratio
Sampling floor: sample κ_4 against the teacher	0.938 / 0.949 / 0.963	1.00 / 0.99 / 1.00
Exact one-vertex births + linear carry (telescoping identity, rel. err. 4.8e-6–1.0e-5)	0.938 / 0.949 / 0.963	1.00 / 0.99 / 1.00
Exact one-vertex births	0.937 / 0.919 / 0.932	0.77 / 0.63 / 0.62
K = 3 births with exact inputs	0.926 / 0.893 / 0.918	1.35 / 1.10 / 1.08
Star alone	0.521 / 0.489 / 0.471	−0.84 / −0.88 / −0.82
C·T alone	0.442 / 0.431 / 0.424	0.88 / 0.93 / 0.85
T·T alone	−0.159 / −0.133 / −0.057	−0.04 / −0.05 / −0.05
Star + T·T	0.490 / 0.445 / 0.436	
Star + C·T	**0.846 / 0.835 / 0.824**	0.04 / 0.05 / 0.03
Star + C·T + T·T	0.849 / 0.840 / 0.830	0.00 / 0.00 / −0.02
Same, hub index of Q projected on the top-r eigenvectors of C_m, r = 32 / 64 / 128	0.67–0.71 / 0.74–0.75 / 0.78–0.81	

Reading the table:

- **Star + C·T matches the Gaussian-restart cubic-birth ceiling** of set A (§5.4: 0.848 / 0.799 / 0.817). It stays below the K = 3 births with exact inputs in the same run (0.89–0.93). Over ages 1, 2, 4 and 6 it builds up as 0.57 → 0.71 → 0.82 → 0.85 at L12 (0.43 → 0.65 → 0.80 → 0.84 at L15).
- **The Round-17 C·T kill was too narrow.** C·T alone sits exactly where the hub-pair carrier was killed (0.42–0.44), but it is complementary to the star.
- **T·T is dead.** It is negative alone and adds ≤ 0.006 on top of star + C·T.
- **The readouts carry the pattern, not the level.** The star and C·T levels cancel (sum level −0.02 to 0.05), so the prediction must enter as a *feature* of a rescaled map, as in V55/V56, not as a raw κ_4 diagonal.
- **The single-piece identities hold on the samples.** The exact one-vertex births split into joint cumulants of the linear, quadratic and cubic images of the response. Per age, the quadratic piece Q1 = 4 κ(quad, lin, lin, lin) correlates 0.934–0.974 with C·T. The cubic piece Cu1 = 4 κ(cub, lin, lin, lin) correlates 0.993–0.998 with star + T·T (plus a diagnostic C·K term).

As a feature: oracle end-to-end test (`fable_affine_fits.json`, `fable_affine_e2e.json`). At each layer 7–16 an affine map

$\kappa_4/v^2 \approx a \cdot \kappa_4\text{^chain}/v^2 + b \cdot \text{pred}/v^2 + c$

(three numbers per layer) is fitted by least squares to MLP #0's teacher, with pred taken from the sample-exact ceiling readouts. The fitted values replace the κ_4 diagonal in the plain chain's mean closure on the same network. **This is an oracle in two ways:** the map is fitted to the teacher on the network it is evaluated on, and the feature's FLOPs are not metered (util stays 0.2476).

κ_4 diagonal in the mean closure, layers 7–16	Fit corr, L7–L16	Final MSE	Ratio
Plain chain (own κ_4)	0.26–0.40	2.273e-8	1.000
Teacher κ_4 diagonal	—	1.591e-8	0.700
Affine(chain) only: the fair baseline	—	2.467e-8	1.085
Affine(chain, star)	0.37–0.51	2.296e-8	1.010
Affine(chain, star + C·T)	**0.82–0.86**	**1.758e-8**	**0.773**
Affine(chain, star + C·T + T·T)	0.81–0.86	1.801e-8	0.792

C·T recovers (1 − 0.773)/(1 − 0.700) ≈ 75 % of the perfect-κ_4 gain. Against the same-maturity affine baseline (×1.085), star + C·T is ×0.713 and star alone is ×0.931.

Component test: the chain already holds the inputs (`fable_component_2026-10-09.py`). The test uses captures of the plain chain with every source dense. Each of a source's three blocks has three legs, and each of a block's three Sym placements (a "role"; nine per source) pairs one leg at m (the anchor L) with the Hadamard product H of the other two legs at t. Then Q_m(t) = (1/3) Σ_s Σ_roles L_s(m) H_s(t)^T. The legs at t are the legs at m transported by Ẏ (checked to 1.3e-7 and 8.2e-7).

Against the sample-exact readouts on decisive neurons, the chain's own readouts correlate:

- C·T: 0.993 (pair 14 → 15, age 1) and 0.996 (11 → 15, age 4);
- star: 1.000;
- T·T: 0.991–0.994.

The C·T content is therefore available from the chain's existing κ_3 legs. The obstacle is cost: formed per (source, role, age), Q costs one dense n^3 product each.

Age-collapse identity (derived; measured exact). Write G_m = diag(g_m)(Ȳ_{m→t} ∘ X_{m→t}), an n × n hub-by-target matrix. Then

ct_i = 8 Σ_m Σ_s Σ_roles Σ_a (G_m)_ai (L_s(m) H_s(t)^T)_ai (8 = 24/3).

Let a_s be the first window layer that holds source s. Its anchor legs are transported, L_s(m) = (Ystep_{a_s} ⋯ Ystep_{m−1})^T L_s(a_s), so the sum over ages collapses onto one matrix per anchor layer:

ct_i = 8 Σ_s Σ_roles Σ_h (L_s(a_s)^T Γ_{a_s})*hi H_s(t)_ih, Γ_a = G_a + Ystep_a Γ*{a+1}.

Γ is built once per target by a backward recursion over the window. On the captures, the collapsed form matches the per-age route to a relative error of 2.9–3.0e-7 at L12 / L15 / L16 (`acds_ct_probe.json`) and 2.9–3.9e-7 at L7–L16 (`acds2_ct_probe.json`), which is float32 rounding.

Anchor grouping (derived; measured exact). The nine anchor legs of a source are linear in its two dense legs, A_s and P_s, plus thin legs (the low-rank feedback and residual columns). Grouping the roles by dense anchor gives

ct_i = 8 Σ_s [Σ_h (A_s^T Γ_{a_s})*hi (H_A,s)_ih + Σ_h (P_s^T Γ*{a_s})_hi (H_P,s)_ih + thin_s,i],

where H_A,s and H_P,s collect the target Hadamards of all roles anchored on A_s or on P_s. The thin_s,i terms are evaluated exactly through the thin products Z_s^T Γ. The grouped form matches the per-age route to 2.6–3.2e-7 at every target L7–L16. What remains is **two dense products per source and target**, plus building G_m and Γ.

Sketches. The dense products are replaced by $A^T \Gamma_a \approx (A^T \tilde{S}a)(S_a{}^T \Gamma_a)$, *with* $\tilde{S} = S\,(S^TS)^{-1}$. $S_a{}^T \Gamma_a = \Sigma\{m \geq a\}\, S_m{}^T G_m$ when the rank-r basis S is transported like a leg. Bases tested:

- "eig": the top-r eigenvectors of C at the window start;
- "rad": C Ω with a Rademacher Ω;
- "direct-Γ": an orthonormal basis of Γ_a Ω per anchor, optionally with target-aware column weights $\omega_i = dec_i \sqrt{\lambda_i} / v_i^2$ and one power step.

The captures hold dense source stacks only at a few layers, so these tests use the consistent source subset: sources born between stack layers are absent until the next stack. On that subset, the exact chain C·T correlates 0.82–0.96 with the sample-exact one, against 0.993–0.996 for the full source set in the component test.

C·T variant	Fidelity to the exact chain C·T (decisive neurons)	Teacher-metric corr of affine(chain, star + C·T), L12 / L15 / L16 or their range
Sample-exact (ceiling)	—	0.859 / 0.858 / 0.843
Exact chain C·T, source subset	1	0.725 / 0.760 / 0.778
Round 1 (all anchors sketched), eig / rad, r = 64	0.93–0.94 / 0.89–0.93	0.60–0.63 / 0.60–0.61
Round 1, eig / rad, r = 128	0.95–0.97 / 0.94–0.96	0.67 / 0.65–0.66
Round 1, eig / rad, r = 256	0.97–0.98 / 0.97–0.98	0.71 / 0.70–0.72
Round 2 (grouped, low-rank anchors exact), eig r = 64	0.91–0.97 (L7–L16)	0.659 / 0.665 / 0.674
Round 2, eig r = 128	0.955–0.983	0.691 / 0.700 / 0.715
Round 2, direct-Γ weighted + power step, r = 64	0.89–0.94	0.660 / 0.672 / 0.674
Round 2, direct-Γ, r = 64	0.87–0.92	0.613 / 0.539 / 0.590

Two lessons:

- **Once the low-rank anchors are exact, the plain eigen-sketch of C is the best basis we tried.** At r = 64 the best direct-Γ basis ties it on the teacher metric at L12–L16, has lower fidelity and a worse end-to-end ratio (×0.885 against ×0.865, below), and costs more. The probe's thin-product count over targets 7–16 is 37.6–51.3 u against 27.6 u.
- **High fidelity is not enough.** In round 2 at L12–L16, sketches that correlate 0.96–0.98 with the exact chain C·T still lose 0.03–0.10 of teacher-metric correlation. The information C·T adds lies in the part the sketch drops.

End to end with the chain's own C·T (`acds_e2e.json`). This uses the same oracle affine map and protocol as above, with the chain's own star and the C·T variant from the captures:

C·T input to the affine feature	Final MSE	Ratio
Teacher κ_4 diagonal (reference)	1.591e-8	0.700
Sample-exact C·T (ceiling)	1.758e-8	0.773
Exact chain C·T, source subset	1.863e-8	0.819
Eigen-sketch, r = 128	1.917e-8	0.843
Eigen-sketch, r = 64	1.966e-8	0.865
Direct-Γ weighted + power step, r = 64	2.012e-8	0.885

The gap from ×0.773 to ×0.819 is at least partly the capture subset, since the full-source component fidelity is 0.993–0.996. That share has not been measured separately.

Layer windows (10 Oct, `results/summaries/oct10/ctwin_{1,2}.json`). The same oracle affine feature is injected only at the target layers of a window. Elsewhere the chain keeps its own κ_4. The base is 2.2732e-8.

C·T input to the affine feature	Window	Final MSE	Ratio
Sample-exact (ceiling, above)	L7–16	1.758e-8	0.773
Sample-exact	L11–16	1.811e-8	0.797
Sample-exact	L13–16	1.951e-8	0.858
Sample-exact	L15–16	2.135e-8	0.939
Sample-exact	L7–10	2.235e-8	0.983
Eigen-sketch, r = 64 (above)	L7–16	1.966e-8	0.865
Eigen-sketch, r = 64	L11–16	2.074e-8	0.912
Eigen-sketch, r = 64	L13–16	2.164e-8	0.952

L11–16 carries (1 − 0.797)/(1 − 0.773) ≈ 89 % of the ceiling's gain, and L7–10 alone is nearly worthless (×0.983). With the r = 64 sketch, however, L11–16 keeps only (1 − 0.912)/(1 − 0.865) ≈ 65 % of the gain, while it holds 72 % of the probe's thin-product count (derived from `acds3_ct_probe.json`; the later targets hold more sources). Cutting target layers therefore saves less than it loses. Cutting rank is the better cut (round 3, below).

Round 3: cheaper bases and source subsets (10 Oct, `acds3_ct_probe.json`, `acds_e2e3.json`). The captures, protocol and oracle map are those of round 2. The new variants are:

- "rsi": a randomized subspace iteration on C at the window start (q = 2, three thin products with C), transported like the eigen-basis;
- "top-6": only the six sources with the largest ‖A_s‖‖H_A,s‖ + ‖P_s‖‖H_P,s‖ are kept, and the others are dropped;
- "young only" / "old only": only the sources anchored after / at the window start are kept.

The cost column is the probe's thin-product count summed over targets 7–16. It excludes forming G_m and Γ.

C·T variant	Thin-product count	Teacher-metric corr of affine(chain, star + C·T), L12 / L15 / L16	Final MSE	Ratio
Eig, r = 64 (round 2)	27.6 u	0.659 / 0.665 / 0.674	1.966e-8	0.865
rsi, r = 64	27.6 + 1.9 u	0.659 / 0.654 / 0.668	1.965e-8	0.864
Eig, r = 48	20.7 u	0.643 / 0.661 / 0.659	1.993e-8	0.877
Eig, r = 32	**13.8 u**	0.620 / 0.629 / 0.637	**1.987e-8**	**0.874**
Eig, r = 64, top-6 sources	20.9 u	0.665 / 0.625 / 0.642	2.063e-8	0.907
Eig, r = 32, top-6 sources	10.4 u	0.631 / 0.613 / 0.610	2.076e-8	0.913
Eig, r = 64, young sources only	—	0.675 / 0.660 / 0.694	2.051e-8	0.902
Eig, r = 64, old sources only	—	0.503 / 0.474 / 0.391	—	—

On every target the age-collapse and grouping identities again hold to 2.6–3.9e-7.

Reading the table:

- **Rank is cheap to cut.** r = 32 keeps ×0.874 against ×0.865 at r = 64, for half the thin-product count. r = 48 is no better than r = 32 (×0.877).
- **rsi ties the eigen-basis** (×0.864) and adds 1.9 u.
- **Dropping sources costs more than dropping rank.** Top-6 gives ×0.907–0.913 and young-only ×0.902. The old-anchored sources carry signal that no cheap subset keeps.
- **The three-layer teacher metric does not rank sketches end to end.** Young-only has the highest correlation at L12 / L15 / L16 (0.66–0.69) but ends at ×0.902, behind eig r = 32 (0.62–0.64, ×0.874). Sketch choices must be made on the end-to-end mean.

Cost and verdict. The score is MSE × max(0.1, FLOPs/2^{41}). V56 uses about 190 u (util 0.186), and our estimate for a deployable eigen-sketch C·T at r = 64 is about 40 u (estimate). Adding 40 u multiplies util by about 1.21, so a raw gain of ×0.865 gives a score of about ×1.05, i.e. roughly neutral. Break-even at ×0.865 needs less than about 30 u (estimate). The probe's count of thin sketch products alone already sums to 27.6 u over targets 7–16 at r = 64, before G_m and Γ are formed.

Round 3 changes this arithmetic. Scaling the 40 u estimate by the probe's thin-product counts (13.8 against 27.6 u) puts eig r = 32 at about 18–20 u (estimate). Util then rises by a factor of about 1.09–1.11, and the score ratio is about 0.874 × 1.10 ≈ 0.96–0.97, i.e. a gain of 3–4 % (estimate). The open risk is the metered op count, which no prototype has measured yet.

Moreover, every ratio in this subsection was measured:

- on one network;
- with an in-sample teacher-fitted map;
- on the plain chain rather than on V56, which already contains the star and the V55 features.

The increment on V56 with an offline-fitted map is **(untested)**.

Conclusion. The κ_4 C·T channel is real. Its ceiling is 0.82–0.85 together with the star, it is worth ×0.773 under an oracle map, and the chain's own legs reproduce it at 0.993–0.996. At r = 64 it is **not affordable**. At r = 32

(round 3) it is at the edge of affordability, at an estimated 3–4 % score gain, with every caveat above still applying. Per the joint oracle (§4.8), κ_3 is the larger lever: ×0.607 against ×0.686 on MLP #0, and ×0.716 against ×0.885 on the synthetic net.

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6. Negative results

Unless marked, ratios are final-MSE or score ratios against the chain, given as MLP #0 / synthetic net. "corr" is the decisive-neuron κ_4 pattern correlation. "Gate" is the pre-registered pass criterion.

The kill types are:

- **C:** a ceiling or oracle (an upper bound);
- **P:** a teacher-forced component test;
- **E:** an end-to-end prototype;
- **K:** a knob.

6.1 κ_3 representation and cost architectures

Idea	Measured (gate)	Kill
Low-rank Hadamard products of source legs	A∘P needs 681–778 of 1024 directions for 99 % energy (flat spectrum)	C
Old content only in the last k layers	Keep last 1/2/3 layers: 4.85 / 4.17 / 3.16e-7 against a chain error of 2.1e-8; dropping one birth: 1.3–2.7e-7	C
Query surrogates: trace, diagonal Hermite, TensorSketch	Error at L14: 98.2 %, 95.4 %, 790 % (d = 2048)	C/P
Pooled or aggregate states: vector metric, Tucker/TT, CP	Rank 32/128/256 → 35 / 7.8 / 1.6 %; k = 128 → 20 %; CP rank 4608 → 1.07e-7	C
Slice-to-slice recursion	R^2 0.030 / 0.047; RSC D21 error 35.9 % vs the chain's 6.6 %	C
Cubic source-index coding	388 / 527 % error (q = 1); 83 / 130 % (32 channels)	P
Birth operator ≈ diagonal + rank k	73 / 57 % (k = 32); 22 / 16 % (k = 384)	C
Adjoint low-rank response	Low-rank path carries 14 % of the response	C
ISDF skeleton / landmarks	56–71 % (anchor), 79–94 % (transported)	P
Birth-column screening	−5.6 % cost for +25–30 % raw	K
Asymmetric contraction rank	Util +0.8 %, raw +2.9 %	E
Young tier at half cost	193 products in every order (§2.6)	count
Explicit κ_3 tensor / lazy circuit	≈ 1567 u per layer / ≈ 245 u against 249 u	count
Short age memory: age modes, shift carrier	h = 10 still 16 %; 6 modes 8.5 %; shift error 2.9 % (gate 2 %)	C
Source-free dressed hypertree (DHMP)	Next-layer D21 error 36–40 % (gate 2.5 %)	P
Diagonal-only κ_3/κ_4, diagonal + low-rank C	Next-d3 error 83–99 %; diagonal + rank-64 C 12–33 %	P
Deterministic moment-matched particles	D21 error 74–160 % against the chain's 6.6 %	E
Response-cashout propagator (use each source once)	Equals keep-last-1: +4.85e-7, about 23× the chain error	C
Richardson extrapolation in rank	Exponent 1.46 vs 2.07; every extrapolation worse	C
Strassen level 6 / 7	8.13 GB against the 8 GB cap; level 7 runs out of memory	cap
Compensating closure with no old tier	7.43e-7 (32×); every knob 6.8–9.9e-7	C
Hermite-tree κ_3 replacing the chain (§5.6)	Closure 10–20× worse	P

6.2 κ_4 channel

Idea	Measured (gate)	Kill
Per-row λ fitted to the (3,1) slice	2.2911 → 2.2988e-8 (worse)	K
Depth multiplier, pairwise transport, 13-feature regression	×1.25 / ×1.16; corr 0.24–0.59	K/P
Young κ_4 carrier, h ≤ 3	G4-weighted error 0.188 / 0.173 (ideal)	C
Quartic response kernel	−6.4 / −2.6 % at ≈ 42 u; not re-tuned	E (parked)
K_3×C all-distinct term	−8.8 % (MLP #0) at ≈ 140 u; cheap form −4.2 % at ≈ 13 u, synthetic run unfinished	E (parked)
Nine (2,1,1) hypertree classes	$R^2 \le 0.07$	P
Carried pair-response modes	≤ 3 % error cut against a 15 % gate	P
SSC 4-mode closure + drift MLP	corr 0.25–0.33 at L15; held-out $R^2 \le 0.004$ (gate 0.10)	P
Role-split λ	×0.971 / ×0.998 (gate 0.97 on both); graded +0.77 % (#332382)	K
Per-source pair innovations γ, η (scalar form)	×0.92 in sample, ×1.05 out of sample	P
Exchange vertex $3YC^{-1}Y$	corr −0.07 to −0.23; error ×26–95	P
Source-resolved skew energy	+≤ 0.004 corr	C
Every second-order carrier in §5.4	0.03–0.56 (gate 0.70)	C
Augmented dense κ_4 core (504aldo F105)	2.04e-8 vs 2.23e-8 at +3 dense units per source-layer	C (cost)
T·T: κ_3 squared at a cubic vertex (§5.8)	Alone −0.16 to −0.06; adds ≤ 0.006 to star + C·T; oracle E2E ×0.792 against ×0.773 without it	C/E
LKV readouts injected as the raw κ_4 diagonal	Level ratio of the ages 1–6 sum −0.02 to 0.00 (§5.8); usable only as a feature	C
Low-rank C·T sketches, r = 32–256 (eig, Rademacher, direct-Γ)	Teacher-metric corr 0.60–0.63 at r = 64 against 0.73–0.78 exact (round 1); oracle E2E ×0.843–0.885 against ×0.773; ≈ 40 u (estimate) against a ∼30 u break-even. Round 3 reopens it: eig r = 32 gives ×0.874 at an estimated 18–20 u (§5.8)	E (parked, cost; r = 32 reopened)

6.3 Correctors and non-cumulant families

Idea	Measured
Covariance patches	2–3e-6 RMS against an error of 6.5–9e-5
True D21 in the feedback legs only	×0.9999
Low-rank covariance-error correction	Needs rank ≈ 256
Exact old tier	−3.9 % MSE for +19 % cost
Online mean-correction refit	R^2 0.02 (chance level 0.013)
Learned residual: ridge / GBM	×0.997 / ×0.995
Monte-Carlo blends and control variates	Weight ≈ 0.002; surrogate corr 0.19 at L16
Copula, TT-cross (60–500 % at width 32), TAP/AMP, SOS	Dead (§4.7)
ARC reference K3 at width 1024	1523 u, raw 3.7–4.5e-8: 5.9× dearer and 1.7× less accurate than V29

6.4 Outside-AI proposals

We consulted outside models in 18 rounds. Below, "In PDF?" states whether the proposal itself contained a measurement on our data. Every verdict is our own measurement.

Round, author	Proposal	In PDF?	Our verdict
R1–2 (8 models)	Compensating closure without the old tier	No	Dead: 32× worse
R3, GPT Sol / Astra	Source-index coding; low-rank birth operator	No	Dead (§6.1)
R4, GPT Sol	**Spherical-input reparameterisation (from a public repo)**	No	**Shipped (V31), −1.1 % raw**
R5, Sol / Astra	20 families (TAP/AMP, copula, TT-cross, SOS, …)	No	Dead
R7, Astra	Quartic κ_4 kernel; teacher-forced closure diagnostic	No	Kernel parked; the diagnostic shaped §4
R7, Grok	Monte-Carlo surrogate, "40×"	No	Measured ×1.0–2.2
R7, unknown pasted AI	"Closed-form" (1,1,1,1), "30×" MC	No	Wrong formula; claim baseless
R7–8, Fable (internal agent)	Error decomposition, FB scale, $K_3 \times C$	**Yes**	FB 1.25 shipped (#332110)
R9, GPT Sol	DHMP, FHQR-2, QCG ("promotion targets, not predictions")	No	Dead (36–40 %, 16 %, by arithmetic)
R10, several validators	Young-tier contraction search	Yes (counting scripts)	Confirmed 193 products; one answer's 49–55 u claim was an arithmetic error
R11, Copilot	Response-state closure	No	D21 35.9 % even from the true state
R11, SciMaster/Bohrium	Diagonal κ_3/κ_4 + low-rank C, "115 u at 1.18× chain MSE"	**Claimed, no run exists**	d3 error 90–99 %
R11, Astra	Channel-history state	Declined ("no update rule")	—
R12, Opus (internal agent)	Deep κ_4 is carried memory	**Yes**	Basis of §5.2
R13, unsigned	Skew-lift carrier	No	corr 0.03
R13, Astra	Dense pair core	**Yes**, self-rejected	corr −0.35 to −0.22
R14, unsigned	Response-cashout; exchange vertex	No	Dead (§6.1, §6.2)
R15, Astra	Skew energy	**Yes**, reproduced identically	+0.004
R16, unsigned	Signed bilinear quartic memory	Attempted; data host unreachable; declared none	0.33–0.39
R16b, Astra	Pair-innovation carrier	**Yes**, reproduced exactly	0.36–0.44
R17, Claude (ours)	Hub-pair C·T births	Yes	0.43–0.44
R17, unsigned	GHQ-6 hypertree bank	Script, own gate KILL	0.47–0.56
R17, SciMaster	Factorised curvature-birth carrier, "confirms ≥ 0.70, passes gate"	**Claimed; the script crashes** (`KeyError 'D21'`); patched, its own gate prints KILL	0.39 / 0.32 / 0.40
R17, SciMaster	Curvature-source factorised carrier, "exceeds 0.7 (PASS)"	**Claimed, no script or output**	The identical design measured 0.43

Round, author	Proposal	In PDF?	Our verdict
R18, unsigned	**HTQB star + path**	Algebra self-test only (n = 4)	**Borderline alone; star shipped as V56**

Two patterns stand out:

- **Renamed dead families.** Several proposals re-derived a dead family under a new name. The response-cashout propagator is keep-last-1; QCG is an n-term CP; FHQR is a short horizon.
- **Claims without runs.** Three documents reported passing measurements that no run supported.

Including the kill list and the project metric in every prompt is what filtered these out.

6.5 Hypotheses killed on 10 Oct

§4.10 places the remaining κ_3 error in the carried late-layer slice and in the d3 level. Three mechanisms for it, and for the FB 1.25 paradox of §2.4, were tested on 10 Oct. Each is killed as measured.

(a) A missing cubic-response C×κ_3 term ("RLL"; P). The hypothesis, from a 10 Oct outside-AI audit, was that the chain misses half of the term with one cubic and two linear Hermite vertices, at first order in the pre-activation κ_3 T.

Derivation (derived). With the K = 3 response h − E h = l u + q (u^2 − v) + r (u^3 − 3vu) (§5.8), the connected partitions give

$\kappa(R_i, L_j, L_k) = r_i\, l_j\, l_k\, [3\, C_{ij}\, T_{iik} + 3\, C_{ik}\, T_{iij}]$ (+ κ_5, dropped).

The proposal also contained a C_jk T_iii term. That partition is disconnected, because the block {i, i, i} does not meet {j, k}, so it does not belong in the joint cumulant. We tested both the connected form and the form as proposed.

Test (`astra_rll_test.py`). With the teacher's inputs at layer l, we map the term to its image dD on the (2,1) slice at l + 1. We compare dD with the chain's residual E = teacher − chain at l + 1 (the captures of §4.10), off the diagonal and on d3 separately. We report:

- $\beta = \langle E, dD\rangle / |dD|^2$;
- ρ = corr(E, dD);
- r_half = |E − dD/2| / |E|, the residual left after adding half of the term.

The pre-registered kill was $\beta \approx 0$ or negative, or $\rho < 0.1$ at L ≥ 9. Values are given for the transitions l → l + 1, l = 2–15, with l ≥ 9 in brackets:

Form	Part	Size relative to the teacher's slice	ρ	r_half
Connected	Off-diagonal	4.1–9.5 %	−0.04 to +0.07 (−0.04 to +0.02)	1.08–1.42 (1.15–1.31)
Connected	d3	4.5–10.0 %	−0.24 to +0.07 (−0.24 to −0.12)	1.23–2.03
As proposed	Off-diagonal	8.5–13.7 %	−0.06 to +0.02	1.23–1.89
As proposed	d3	8.9–15.1 %	−0.23 to +0.01	1.54–2.95

Verdict: KILL.

- The connected term is comparable in size to the residual itself (4–10 % of the slice, against a residual of 4–7 %), yet it is uncorrelated with the off-diagonal residual ($\rho \approx 0$, β −0.04 to +0.02 at $l \geq 9$).
- On d3 it is anti-correlated at depth.
- Adding half of it makes the off-diagonal slice residual worse, by 15–42 % from L3 on (8 % at L2 → 3), and makes the d3 residual worse by 23–103 %.

This is consistent with the chain already carrying the term in its feedback legs.

(b) A feedback-scale confound ("γ-patch"; E). The same audit proposed the following explanation for the paradox of §2.4. The FB scale multiplies not only the linear K×C coefficient but also a quadratic Sym(Xt, P, Yt) cross term of the thin feedback legs, which grows as FB^2. FB 2.0 would then restore the linear term correctly but overshoot the quadratic one.

The patch (`experiments/variants/v56_fbg.py`) sets the linear coefficient to FB and multiplies only the Xt·Yt contributions by $\gamma = (FB_base/FB)^2$, which holds them at FB_base^2. We ran it on the 100-MLP harness (§8.2) on top of the V56 A-class estimator (v56As_fs, FB 1.25). No MLP failed in any run. Paired ratios are against the base:

Variant	Linear FB	Quadratic Xt·Yt term held at	Raw MSE	Score	Paired geo-mean ratio (90 % CI)	Wins
base (V56 A, shipped)	1.25	1.25^2	2.0194e-8	3.7586e-9	1	—
Unpatched FB 2.0	2.0	2.0^2	2.0689e-8	3.8508e-9	×1.0244 (1.0211–1.0277)	11/100
γ-patch, FB 2.0	2.0	1.25^2	2.0686e-8	3.8522e-9	×1.0243 (1.0209–1.0273)	10/100
γ-patch, FB 2.0	2.0	1.0^2	2.0686e-8	3.8522e-9	×1.0243 (1.0212–1.0274)	11/100
γ-patch, FB 1.75	1.75	1.25^2	2.0403e-8	3.7995e-9	×1.0103 (1.0080–1.0124)	25/100

Verdict: KILL.

- Holding the quadratic term at 1.25^2, at 1.0^2, or letting it grow to 2.0^2 gives the same result (×1.0243–1.0244), so the quadratic cross term is inert.
- The loss follows the linear coefficient: ×1.0103 at 1.75 and ×1.0243 at 2.0. The linear K×C restoration itself hurts the final mean.

The paradox of §2.4 remains open, with this explanation removed.

(c) The $\kappa_4 \to \kappa_3$ feed channel (C/E). A 10 Oct consultation (`consultations/fable_2026-10-10_k3_design.md`) proposed the following. Among the first-order terms of the all-distinct post-ReLU κ_3, the only one whose input the chain lacks is the feed $g_c\ \Phi_i\ \Phi_j\ \kappa_4(z_m)_ccij$ from the pre-activation κ_4 at a quadratic vertex c (derived in the design note). According to the design note's reading of the code, the chain computes this feed from its memoryless κ_4 model (§1.2), whose only pattern is C and which does not depend on the vertex neuron c. The design would replace it with a carried cubic κ_4 (star + C·T, §5.8) in thin legs.

Ceiling (`fable_k3feed_ceiling_2026-10-10.py`; Kaggle CPU in five parts; N = 10^6 exact samples of MLP #0; targets L9–16; `k3feed_mlp0_part{1..5}.json`). The script forms several versions of the feed read on the (2,1) slice at t:

- **exact:** the exact feed of every vertex;
- **bornJ4 / bornJ6:** only the κ_4 born within J layers, which is the ceiling of any carrier with J-layer births;
- **star_J4 / star_J6:** the Gaussian-star-fed feed in closed form.

The sampled C·T part of the design's representable feed was not run. As a check, the feed from the Gaussian layer z_1 sits at its own noise level (0.13–0.18 against noise 0.14–0.20).

Quantity, L9–16	Value
Exact feed size, off-diagonal, relative to |Y21|	22.5–24.2 %
Exact feed size, diagonal, relative to |d3|	23.3–23.6 %
Split-half reliability of the exact feed, off-diagonal / diagonal	0.871–0.966 / 0.939–0.981
Sampling noise of the exact feed, relative to |Y21| (derived)	3.3–6.0 %
bornJ6 against exact, cross-half corr, off-diagonal / diagonal	0.871–0.960 / 0.938–0.976
bornJ4 against exact, same	0.861–0.949 / 0.932–0.969
star_J6 against bornJ6, same	−0.871 to −0.921 / −0.927 to −0.956
star_J6 size relative to |Y21|	12.9–15.9 %

The pre-registered validity sub-gate, a split-half reliability of at least 0.90 at every target, is not met at L9 (0.871) or L10 (0.894).

End to end (`experiments/variants/r20_feed_oracle.py`). The variant removes the chain's own feed at the slice readout in a window and adds a feed read from file. It runs on the plain chain of §4.8, MLP #0. Hook validity: the plain configuration reproduces 2.2732e-8 exactly (`teacher_eval_feedor1.json`). The util of these runs is not comparable with the chain's, because the hook adds a second readout.

Feed in the window	Window	Final MSE	Ratio
None (chain feed removed)	L9–16	1.715e-7	7.54
None	L1–16	2.724e-7	11.98
Exact (sampled)	L9–16	2.535e-8	1.115
bornJ6	L9–16	2.955e-8	1.300
bornJ4	L9–16	4.409e-8	1.940
star_J6	L9–16	4.783e-7	21.0
star_J6, sign flipped	L9–16	4.255e-8	1.872
Chain's own feed × 0.80 / 0.90 / 0.95	L9–16	2.777 / 2.371 / 2.284e-8	1.222 / 1.043 / 1.0045
Chain's own feed × 1.00 (dumped and re-injected)	L9–16	2.2725e-8	0.9997
Chain's own feed × 1.10 / 1.20	L9–16	2.487 / 3.011e-8	1.094 / 1.324

The removal runs are in `teacher_eval_feedor2.json` and the injections in `results/summaries/oct10/feedorE2E_{3,4,5,6}.json`. The pre-registered gates were: exact ≤ 0.80 PASS and > 0.92 KILL; bornJ6 ≤ 0.85 PASS and > 0.95 KILL. For reference, the teacher's κ_3 in L9–16 is worth about ×0.69 (§4.10).

Slice-level attribution (measured; `results/summaries/k3feed/feed_attribution.json`, script `experiments/probes/k3_error_profile.py`).

- At L9–L15, the chain's own feed correlates +0.95 to +0.97 with the exact feed. The exact feed is 12–26 % larger.
- The correction (exact − chain feed) correlates +0.41 to +0.59 with the chain's slice residual.
- Yet replacing the chain's feed with the exact one still increases the slice residual: from 5.2–6.2 % to 6.0–6.8 % of |Y21|.
- The best scalar rescaling of the chain's feed (β = +0.03 to +0.08) lowers the slice residual by only 0.5–3 %.

Reading:

- **The feed is large, and the chain needs it.** It is 22–24 % of |Y21|, and removing it costs ×7.5 (L9–16) or ×12 (all layers).
- **The chain's feed has the right shape and the wrong size**, and the correction points along the residual.
- **Every injection is nevertheless worse than the base, the exact feed included** (×1.115, KILL by its own gate). The chain's feed sits at a sharp optimum at scale 1.0: −5 % costs 0.45 % and ±10 % costs 4.3 % / 9.4 %. Scaling it up by 10 % hurts, although the exact feed is 14–24 % larger.
- **The sampled feed is noisy at the level that matters.** Its sampling noise, 3.3–6.0 % of |Y21|, is comparable to the chain's whole slice residual at L9–16 (about 5–7 %, §4.10). This would explain why the exact feed raises the slice residual although its correction is correlated with it.
- **The Gaussian star has the opposite sign to the sampled born feed** (corr −0.87 to −0.92). We have not established whether this is a sign-convention error in the closed form or a physical sign **[OPEN]**. Neither sign helps: ×21.0 as computed and ×1.872 flipped.

Verdict: KILL as measured. The experiment is decisive about the N = 10^6 estimate of the exact feed, not about the true feed. The ceiling is noise-limited: reliability is 0.87–0.97, and the validity sub-gate failed at L9–10. Bringing the feed noise well below the residual would need about 10× more samples. With noise $\propto 1/\sqrt{N}$, that would put it at about 1.0–1.9 % of |Y21| (estimate). A definitive ceiling, and the sampled C·T part, remain **[OPEN]**.

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7. What a better method must satisfy

7.1 Structural constraints (from §2 and §4–§6)

1. **Keep the κ_3 chain at full fidelity.**
   - The mean closure needs κ_3 to about 1 % (§5.6) and D21 to about 2 % (tolerance law).
   - Cheap κ_3 cannot be bought back with κ_4 (§5.1).
   - The κ_3 memory is ≥ 10 ages long, coherent, and not low-rank (§4.5).
   - No compressed linear κ_3 state is closed under transport (§2.6), and none of the more than 20 κ_3 representations we tried came within 5× of its gate (§6.1).
2. **Carry odd information.** Any closure built from even statistics of the weights loses the deep κ_3 entirely (§2.2).
3. **Deliver the κ_4 pattern, not the level, on decisive neurons.**
   - The pattern alone is worth 108 % of the κ_4-oracle gain.
   - Reaching correlation ≥ 0.70 at L12 / L15 / L16 requires births that contain **both the quadratic and the cubic Hermite response** of each ReLU.

- These births must cover at least ages 1–4, and ideally 1–6 (§5.4, set C).
- Second-order births of any form are capped near 0.45–0.56.
- The V56 star plus a κ_3-dressed C·T readout meets this (ceiling 0.82–0.85), and the chain's own legs already carry its inputs (§5.8). It must enter as a feature of a rescaled map, because its level is near zero.

4. **Improve the κ_3 input first.**
   - The teacher's κ_3 slice and d3 are worth more than the κ_4 diagonal on both networks (×0.607 against ×0.686, and ×0.716 against ×0.885), and the two together give ×0.272 (§4.8).
   - A new κ_3 representation needs at least the cubic response of each ReLU: quadratic births make 10–21 % D21 and d3 error in one step, against a 2 % tolerance (§4.9).
   - The target is the carried late-layer error: net of teacher noise, 5–7 % in the slice at L8–16 and a 1.4 % shrink of the d3 level at L16. Teacher κ_3 helps only at L9–16 (§4.10). A missing C×κ_3 term, the feedback scale and the κ_4→κ_3 feed (at N = 10^6) are ruled out as the cause (§6.5).

7.2 Budget constraints

- **Ops and memory.** At most about 1,000 extra warm-call ops above the B-class base, which keeps the total ≤ ~18k for nomination. Total memory under 8 GB, measured under a subprocess runner. No host synchronisations.

- **FLOPs: break-even arithmetic.** A carrier worth 74 % of the oracle gain would cut raw to about ×0.78 (estimate, from the 50 %-blend oracle). On our ≈ 192 u:

  - at +30 u, the score improves by about 10 % (estimate);
  - at +60 u, it is about 2 % worse (estimate).

  So the carrier must cost ≤ ~30 u. - **Starting point.** The V56 star already supplies part of the cubic signal at 0 u. A new carrier is worth only what it adds on top of V55 + star: 0.65–0.74 leave-one-layer-out, measured with teacher μ and C on one network.

7.3 Score targets (raw × util arithmetic)

Target	Requirement
3.0e-9	κ_4-oracle raw (1.61e-8) at today's util of ≈ 0.187, or today's raw (1.95e-8) at util 0.154 (≈ 158 u)
Rank 12 on the 8–9 Oct board (2.2e-9)	raw 1.18e-8 at util 0.187, or 1.47e-8 at 0.15
Leader class	raw 1.4–1.76e-8 at util 0.10–0.16

Cost alone cannot get there. Today's raw at the 0.1 floor would give 1.95e-9. That needs a 47 % cost cut, which §2.6 and §6.1 show is unavailable inside the sum-of-sources family.

The 1024-wide leaders therefore pair a raw MSE 1.2–1.35× lower with a cost 1.25–1.9× lower. Their mechanisms are not public.

7.4 Open design target [OPEN]

The design target is **a factorised κ_4 carrier whose births come from the cubic (and quadratic) Hermite response of each ReLU over ages 1–6, read through the existing κ_3 source legs, for ≤ 30 u and ≤ ~1,000 ops.**

Its ceiling is 0.80–0.85 (set C, ages 1–6) and 0.68–0.74 for ages 1–4. The value of reaching the oracle is about 3.0e-9 at the current cost (estimate).

Status on 10 Oct (§5.8). The readout side of this target now exists:

- star + C·T has a ceiling of 0.82–0.85, and the chain's own κ_3 legs reproduce C·T with fidelity 0.993–0.996;
- through an oracle affine map it is worth ×0.773 end to end;
- the age-collapse identity and A/P anchor grouping make it exact at two dense products per source and target.

The missing piece is cost:

- the eigen-sketch at r = 64 keeps ×0.865 at about 40 u (estimate), which is roughly score-neutral on V56's ≈ 190 u; at r = 128 it keeps ×0.843, with twice the probe's thin-product count (55.3 u against 27.6 u over targets 7–16);
- break-even needs ≤ ∼30 u at that accuracy, or more accuracy at the same cost;
- round 3 (§5.8) found such a point: eig r = 32 keeps ×0.874 at an estimated 18–20 u, about a 3–4 % score gain (estimate), with the metered op count still unmeasured.

No estimator implementation exists yet, and the increment on top of V56, with a map fitted offline and transferred, is untested. In parallel, the joint oracle (§4.8) makes a better κ_3 input the larger open lever. Its first candidate mechanism, the $\kappa_4 \to \kappa_3$ feed, is killed at $N = 10^6$ but is not ruled out at larger N (§6.5).

Smaller open items:

- the cheap $K_3 \times C$ term on the synthetic network (−4.2 % on MLP #0 at ≈ 13 u, unfinished);
- the HTQB path term end to end (+0.04–0.06 leave-one-layer-out, untested in the chain);
- the quartic kernel re-tuned jointly with λ and FB.

7.5 A lesson: local corrections against coupled errors

Again and again in this project, a change that makes one channel locally more correct made the final mean worse. All of the following were measured:

- **Full FB restoration** (§2.4) brings the slice closer to the derivation at L4–10, but the final mean gets 3.4 % / 3.1 % worse on the teacher pair. FB 2.0 is ×1.024 on 100 MLPs (§6.5).
- **The γ-patch** (§6.5), which keeps the linear restoration and removes its quadratic side effect, is no better (×1.0243).
- **Scaling the chain's $\kappa_4 \to \kappa_3$ feed** (§6.5) by ±5–10 % in either direction costs 0.45–9.4 %. This holds even upward, toward the exact feed, which is 12–26 % larger and whose correction correlates with the residual. Injecting the sampled exact feed gives ×1.115.
- **The analytic RLL term** (§6.5), of the right order and comparable size, makes the slice residual 15–42 % worse when half of it is added.
- **Single κ_4 inputs:** injected alone, the off-diagonal κ_4 slices give ×1.03–1.22 (§4.2), and the chain's κ_4 pattern at the teacher's level gives +21.5 % (§5.1).

Only **joint** input oracles give large gains. The teacher's κ_3 and κ_4 together give ×0.272, against ×0.607 and ×0.686 alone (§4.8), and adding the variance takes the ratio to the noise floor (×0.057, §4.2).

The pattern is consistent with the chain's channel errors partly cancelling in the mean closure: correcting one channel uncovers the errors it was offsetting in the others. We have not measured this cancellation directly

[OPEN]. In the feed case, Monte-Carlo noise in the injected quantity is a competing explanation.

The practical lesson for moment-chain estimators is narrower, and it rests only on the measurements above:

- a local test (a slice residual, a one-step error, a pattern correlation) is a necessary gate, but it is not a sufficient one;
- candidates must be promoted on the end-to-end mean;
- a single-channel correction should be expected to need joint re-tuning of the channels it is coupled to (λ, the FB scale, the feed scale).

FB 1.25 is the only correction in this list that gained (§2.4), and it did so at a quarter of the derived restoration.

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8. Reproducibility

8.1 Versions and data

- Software: whestbench 0.16.1, flopscope 0.12.1.
- Dataset revision: v2-phase2.
- Every graded number in this paper carries its AIcrowd submission id. The shipped estimators are in `submissions/<variant>/estimator.py`, each with a lineage header naming every change; for example, `submissions/v56As_fs/estimator.py` (#334870) and `submissions/v56s_fs/estimator.py` (#334869).

8.2 Harness

The 100-MLP paired harness runs on the public mini split with its 10^9-sample truth, using Kaggle CPU runners A–E (`kaggle/runners/`). It is compared with `experiments/probes/kaggle_pair_compare.py`, which reports:

- the geometric-mean ratio;
- wins;
- a bootstrap 90 % interval;
- the score MSE × max(0.1, util);
- the residual profile.

The base is bit-identical across runners, and paired standard errors are 0.01–0.15 %.

Calibration against the grader:

Change	Harness	Graded
V55 B / A	Projected 3.8764e-9 / 3.8562e-9	3.8762e-9 / 3.8582e-9 (within 0.05 %)
Transport join	×0.959	−3.4 %
R_OLD 352	×0.9930	−1.1 %
v52 safe B / A vs v51	+2.1 % / +0.9 %	+1.9 % / +0.8 %
V56 synthetic-fit map	×0.9964	−0.29 % (B) / −0.21 % (A)
Star ×1.25	×0.9972 additional	No additional gain

Every predicted sign matched the grade. For single changes worth more than about 1 %, the predicted size was accurate to within 0.05–0.7 percentage points. For gains under 0.5 % the size was unreliable: the graded

gain came in at roughly 0 to 2× the predicted one.

The harness measures residual time but does not enforce the cap. Its in-process mode did not reproduce the V56 memory failure; only `whest run --runner subprocess` did (§3.4).

8.3 Scripts and result files

Result files are in `results/summaries/` unless a path is given.

Section	Scripts → results
§2.5 identities	`experiments/probes/r17_curvature_identities.py` → `r17_curvature_identities.json`
§3.1 ledger	`experiments/probes/m1_matmul_ledger.py` → `m1_matmul_ledger.json`, `m1_matmul_ledger_callers.json`
§4 error map	Teacher scripts in `kaggle/k4/` (pre- and post-ReLU teachers), the error-map probes listed in `docs/build_notes.py`, and `opus12_errmap.py`
§4.8 joint oracle	`experiments/variants/r20_joint_oracle.py` → `r20_joint_oracle.json`
§4.9 one-step channels	`experiments/probes/r20_onestep_channels.py` → `r20_onestep_channels.json`
§4.10 κ_3 error profile and windows	`experiments/probes/k3_error_profile.py` → `chain_k3_error_profile.json`, `teacher_k3_noise_estimate.json` (and the §6.5 (c) slice attribution `k3feed/feed_attribution.json`); κ_3 windows through `r20_joint_oracle.py` (Kaggle wrappers `kaggle/k3loc/esfer_{1,2,3}/run.py`) → `oct10/k3loc_{1,2,3}.json`
§5.1 oracles	`experiments/chain/k4_oracle_run2.py` → `k4oracle2.json`, `k4oracle2_joint_*.json`, `k4oracle2_target_*.json`
§5.2 memory	`opus12_k4_carry.py`, `q18_k4_horizon.py`, `opus12_q18_noise.py`
§5.3 V55	`experiments/variants/v55_k4c.py`, `experiments/probes/k4_pattern_features.py` → `k4c_fit.json`, `k4_pattern_features*.json`, `teacher_eval_v55_k4c.json`, `kaggle_v55_deep.txt`, `kaggle_v55_sweep1.txt`, `kaggle_v55_sweep2.txt`
§5.4 ceilings	`r16_sbqm_ceiling.py`, `r16b_astra/`, `experiments/probes/r17_hermite_family_ceiling.py` → `data/kaggle_out/r17_ceiling/r17_hermite_family_ceiling.json`, `r17_hermite_family_ceiling_B.json`, `data/kaggle_out/r17_ceiling_c/r17_hermite_family_ceiling_C.json`
§5.5 HTQB and V56	`r18_htqb/`, `r18_htqb_combo.py` → `round18_htqb.json`, `r18_htqb_combo.json`; `experiments/variants/v56_star.py` → `k4s_fit.json`, `kaggle_v56_tuning.txt`
§5.6 tree test	`r19_tree_ceiling.py`, `r19_tree_rescale.py` → `r19_tree_ceiling.json`, `r19_tree_rescale.json`
§5.8 C·T channel	`fable_ceiling_2026-10-09.py` → `fable_ceiling_2026-10-09.json`; oracle affine fits and E2E through `r20_joint_oracle.py` (Kaggle wrapper `kaggle/fable_e2e2/kernel/run.py`) → `fable_affine_fits.json`, `fable_affine_e2e.json`; `fable_component_2026-10-09.py` → `fable_component_2026-10-09.json`; `acds_ct_probe.py`, `acds2_ct_probe.py` → `acds_ct_probe.json`, `acds2_ct_probe.json`, `acds_e2e.json`; layer windows (Kaggle wrappers `kaggle/acds/ctwin_{1,2}/run.py`) → `oct10/ctwin_{1,2}.json`; round 3: `acds2_ct_probe.py --lite 1 --ranks 32,48,64` (wrapper `kaggle/acds/kernel/run.py`) → `acds3_ct_probe.json`, E2E (wrapper `kaggle/acds/e2e3_kernel/run.py`) → `acds_e2e3.json`
§6.5 (a) RLL	`experiments/probes/astra_rll_test.py` → `astra_rll_test.json`
§6.5 (b) γ-patch	`experiments/variants/v56_fbg.py` (runner sets `kaggle/fbg/set_{1..5}`) → `kaggle_fbg_100mlp.json`; per-MLP outputs `data/kaggle_out/fbg_{1..5}/`, compared with `kaggle_pair_compare.py --ref base_fbg`
§6.5 (c) $\kappa_4 \to \kappa_3$ feed	Design `consultations/fable_2026-10-10_k3_design.md`; ceiling `experiments/probes/fable_k3feed_ceiling_2026-10-10.py` (CPU wrappers `kaggle/k3feed/cpu_{1..5}/`) → `k3feed/k3feed_mlp0_part{1..5}.json`; E2E

Section	Scripts → results
	`experiments/variants/r20_feed_oracle.py` (wrappers `kaggle/feedor/k1–k6`) → `k3feed/teacher_eval_feedor1.json` (hook validity), `k3feed/teacher_eval_feedor2.json` (removal), `oct10/feedorE2E_{3,4,5,6}.json` (injections)

The kill register is `docs/architecture_research/KILL_AUDIT.md` (27 Sep, 74 entries). The round verdicts are `consultations/round14_verdicts.md` through `round18_verdicts.md` and `consultations/new_architecture_step1.md`. The experiment queue is `docs/experiment_queue/queue.json`.

Unless a path is given, probe scripts are in `experiments/probes/`. All scripts will be released under MIT with the write-up after the competition closes.

8.4 Known inconsistencies in our records

- The κ_4-diagonal oracle appears as 1.600e-8 and 1.609e-8 in two runs.
- The κ_4-diagonal oracle ratio is ×0.702 in §4.2 (base 2.290e-8), ×0.686 in §4.8 (base 2.273e-8, every layer) and ×0.700 in §5.8 (same base, layers 7–16 only). The bases and injection windows differ; none of the three is a typo.
- The chain's κ_4 correlation is 0.78–0.83 over all neurons but 0.26–0.34 on decisive neurons. Both figures are correct for their populations.
- The grade of #332395 (last-block λ ×1.25) is not recorded in our files.
- The id of the failed v54A submission (7 Oct) is not recorded in our files.
- The chain's slice error is "6.7 % at L15" in §4.5 (base 2.290e-8) and 6.6 % at L15 in §4.10 (the §5.8 captures). Its d3 error is 3.2 / 3.8 / 4.6 / 4.7 % at L7 / L11 / L15 / L16 in §4.3 and 3.0 / 3.6 / 4.4 / 4.6 % in `chain_k3_error_profile.json`. The two come from different chain runs; both include teacher noise.
- Our working notes priced eig r = 32 at "≈ 18 u". The probe's thin-product count is 13.8 u (`acds3_ct_probe.json`), which excludes G_m and Γ. §5.8 therefore quotes 18–20 u as an estimate, scaled from the 40 u estimate for r = 64.
- The slice-level feed attribution of §6.5 (c) was first quoted for L10–L14 only (+0.96, 14–24 %, +0.45 to +0.58). The saved file `k3feed/feed_attribution.json` covers L9–L15: +0.95 to +0.97, 12–26 %, +0.41 to +0.59.
- The feed-oracle runs report util 0.3108–0.3425, because the hook performs a second slice readout. These values are not comparable with the chain's 0.2476.

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9. Disclosure of LLM use

Primary assistant. The main engineering and research assistant was Claude Code (Claude Opus 5 and 5.5), directed by the participant. It wrote and ran most of the probes, implemented the estimator variants, and maintained the notes. Internal Claude subagents did specific measurement tasks; every number they reported was measured. These subagents were:

- Fable 5 (Rounds 7–8: the all-distinct κ_3 decomposition, the FB scale and the K_3×C term);
- Opus (Round 12: κ_4 as carried memory);
- Sonnet (Round 10: the young-tier count).

Outside models consulted. We sent prompts to:

- GPT-5.6 Sol and Terra;

- Astra 6;
- Gemini 3.1 Pro, Grok, DeepSeek and Copilot;
- SciMaster/Bohrium;
- several unsigned answers.

Origin of shipped ideas.

Shipped idea	Origin
Spherical-input reparameterisation	GPT Sol R4, pointing to a public repository
HTQB star	Round-18 proposal, built on our Round-17 ceilings
Parity argument	Fable 5 consultation

Verification. Every outside claim was re-derived or re-measured by us before entering this paper. Several claimed measurements were found to be unsupported, and §6.4 says which. No leaderboard feedback entered any fitted coefficient. The V55 and V56 maps were fitted only on MLP #0 and the synthetic network against our own Monte-Carlo teachers.

Licensing and attribution. The baseline estimator is 504aldo's V29, used under the MIT licence with its copyright header and LICENSE retained. All other changes are ours.

References

1. Wu et al., arXiv:2605.05179 (title to be filled in from the arXiv entry) (the K3 "simple" chain, §4.3, and the reference implementations used in §6.3).
2. 504aldo, *whest-p2-cumulant-k3* (GitHub, MIT licence): the V29 factorised third-cumulant chain with memoryless κ_4 regeneration, D21 feedback, an age-gated shared basis and pooled Strassen–Winograd. Forum post #18218. Finding F105 on the r = 1 κ_4 core.
3. WhestBench Phase 2 forum threads:
    - **18218 (504aldo chain);**
    - **18216 (yujigon, zero-constant closure with CP κ_3);**
    - **18214 (venture_agent, exact bivariate ReLU covariance);**
    - **18200 (baghyeonsig, blended MC);**
    - **18219 (nkosi_ndwandwe, MC economics);**
    - **18199 (keenanpepper, no heavy tail across MLPs);**

- **18175 (Puffi, Phase 1 SSC stack).**

4. barnobarno666, public Phase-2 repository, Phase 8 (spherical-input reparameterisation).
5. Organiser ruling of 11 Sep 2026 that Strassen–Winograd is legal at any recursion depth (competition forum).